

Residential Building / Additional Dwelling Unit (New) Permit

Permit Set 1645 ST LUKE RD P# 2023 004477 CPBC R# 3739010310134020000

PERMIT INFORMATION

Address	Permit number	Date issued
1645 ST LUKE RD	2023 004477 CPBC	2023-07-21

REVIEWED BY

If you have any questions regarding the review of these drawings please contact:

Architectural plans review	Zoning review
Marwan Al-Ezzi	Clarke Gallie
malezzi@citywindsor.ca	info@nccapital.ca

INSTRUCTION AND ATTENTION

It is the responsibility of the Applicant to print **full** size the entire approved permit package and provide at the time of inspection.**All Engineered product shop drawings shall be stamped by a licensed Professional Engineer of Ontario.**

Shop drawings will be reviewed for approval by a Building Official.

Products requiring the submission of shop drawings include but not limited to, roof trusses, floor systems and wood beams.

Prior to the start of framing, email a stamped PDF copy of the shop drawings to buildingdept@citywindsor.ca. Email to include Permit Number and Property Address for reference.

Hard copy of shop drawings to be provided to the Building Inspector at the time of framing inspection.

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THE CORPORATION OF THE CITY OF WINDSOR
PLANNING AND BUILDING DEPARTMENT

John Revell, MCIP, RPP
Chief Building Official



Residential Building / Additional Dwelling Unit (New) Permit
Permit Set 1645 ST LUKE RD P# 2023 004477 CPBC R# 3739010310134020000

LOCATION OF WORK

Address	Zoning	Legal description
1645 ST LUKE RD	RD1.3	PLAN 1028; LOT 4

PERMIT INFORMATION

Permit number	Date applied	Date issued
2023 004477 CPBC	2023-01-25	2023-07-21
Status		
Issued		
Issuer		
Marwan Al-Ezzi		
Reviewed by		
Marwan Al-Ezzi, Christopher Odecki		

GRANTED TO

Applicant, Payer	Builder	Designer
NC Capital	Nate Schaly	Philip Fernandes
Clarke Gallie	416 Brown Crescent	956 McDougall St
416 Brown Cres	Amherstburg, ON N9V0C2	Windsor, Ontario N9A 1M1
Amherstburg, ON N9V 4E1		Individual BCIN 37630
		Company BCIN 47050
Mechanical consultant	Property owner	
Lawrence Mailloux	4052013 CANADA INC.	
511 Cambridge Court	Roger Bramhall	
Tecumseh, ON N8N4B4	416 Stonebrook Blvd	
Individual BCIN 37377	Lakeshore, Ontario N8N 5H2	

CONSTRUCT DETACHED ADDITIONAL DWELLING UNIT AT REAR OF SINGLE FAMILY DWELLING AS PER APPROVED PLANS.

Personal information contained in this form and schedules is collected under the authority of subsection 8(1.1) of the Building Code Act, 1992, and will be used in the administration and enforcement of the Building Code Act, 1992. Questions about this collection can be made to the Freedom of Information Co-ordinator, Council and Customer Services, P.O. 1607, 350 City Hall Square West, Windsor, Ontario N9A 6S1, 519-255-6285

This permit is granted subject to the terms on the Application and on condition that work authorized by the permit shall be completed in all respects in accordance with the plans, specifications and any other information on the basis of which the Permit was including the City of Windsor Zoning Bylaw and other Applicable Law. For Building Permits, Section 8 of Bylaw 11884 requires notice to be given to the Chief Building Official by the permit holder in advance stages of construction specified in Subsection 2.4.5 of the Ontario Building Code and as indicated on the Notice to Owners and Builders attached to the permit drawings.

THIS PERMIT MUST BE POSTED ONSITE. THE PLANS AND SPECIFICATIONS REVIEWED AND ISSUED BY THE BUILDING AND DEVELOPMENT DEPARTMENT MUST ALSO BE KEPT ONSITE AND MADE AVAILABLE TO INSPECTORS AS REQUIRED BY LAW.

INSPECTIONS

Inspections are mandatory for any construction, alterations or demolition for which a building permit is issued. To ensure compliance with structural, health and safety standards established by the Ontario Building Code, inspections are provided by building and mechanical inspectors and are required during all stages of construction. Inspections are also undertaken to ensure that construction is done with the benefit of a building permit.

TO SCHEDULE AN INSPECTION CALL 519-255-6453 (24 HRS NOTICE)

REQUIRED INSPECTIONS

Name	Description
Water service (S)	Installation of new water service. Water must be available and a tracer line is required if the water service is polyethylene.
Sanitary sewer (S)	Installation of sanitary sewer from the property line to the interior of the building. A test for this portion of the plumbing system is required.
Storm sewer (S)	Installation of storm sewer from the property line and around the building. A test for this portion of the plumbing system is required.
Footing (S)	Placement of all footing forms prior to pouring concrete. Geotechnical review may be required to confirm soil suitability.
Foundation wall (S)	Construction of foundation wall including all external elements and perimeter drainage system. (Prior to Backfill)
Plumbing underground (S)	Installation of all plumbing that will be covered by the concrete floor. A test for this portion of the plumbing system is required.
Rough plumbing (S)	Installation of all plumbing including drains, waste, vents and water supply lines. A test for this portion is required including a 100 psi test on all water supply lines.
Rough heating (S)	Installation of all mechanical supply air and return air duct work. Seal all joints in ducts. Installation of hot water heating piping (including in-floor). A test for this system is required.
Framing (S)	Construction of all structural framing and stairs. (Rough Plumbing and Heating passed and doors, windows & roofing installed.) Engineered drawings must be provided for roof trusses.
Insulation (S)	Installation of the insulating material and vapour barrier. Blown-in insulation can be installed after the ceiling drywall is applied and confirmed at the Occupancy inspection.
Rear yard drain (S)	Installation of private drainage system in the rear yard. Typically consists of a catch basin and drainage tile. The required swale will be confirmed at the Exterior Final inspection.
Plumbing final (S)	Installation of all CSA approved plumbing fixtures and hot water tank (water meter must be installed). A test for this portion of the plumbing system may be required.
Heating final (S)	Installation of furnace, air conditioner (if included in the permit) and grills/registers. An Air Balance Report may be required.
Occupancy (S)	Completion of ALL health and safety, Heating and Plumbing components. There must be no outstanding Building Code Orders. (Issuance of an Occupancy permit may be required)
Permit Final (S)	Completion of all required items under the permit including the site grading and the surveyor's final grading certificate.

INSTRUCTIONS

THE PERMIT HOLDER AND/OR PROPERTY OWNER IS RESPONSIBLE FOR ENSURING THAT ALL REQUIRED/APPLICABLE INSPECTIONS ARE SCHEDULED AND THAT THE PERMIT IS COMPLETE

WHAT INFORMATION MUST BE PROVIDED AT THE TIME OF INSPECTION?

- THE PERMIT
- THE APPROVED SET OF PLANS THAT HAVE BEEN INCLUDED WITH YOUR PERMIT PACKAGE
- ANY TEST RECORDS OR RELEVANT DOCUMENTATION
- SAFE ACCESS TO THE SITE AND THE AREAS WHERE INSPECTION ARE TO BE PERFORMED.

SURVEYOR TO VERIFY THAT THE BUILDING SETBACKS DO CONFORM TO CITY OF WINDSOR ZONING BY-LAW 8600 REQUIREMENTS PRIOR TO CONSTRUCTION.

FRONT YARD PAVING AND SURFACING:

For any one family dwelling, semi-detached, duplex dwelling, double duplex dwelling, row dwelling or multiple dwelling containing a maximum of four (4) dwelling units, not more than 50 per cent (50%) of the total surface area of the required front yard shall be covered with any one (1) of or combination of the following: paving block; brick; concrete; asphalt; construction grade aggregate or any similar material.

EAVES

Eaves may extend a maximum of 0.6 metres into a required yard unless the required yard is less than 1.2 metres in width in which case the maximum eave projection shall be 0.3 metres;

ELECTRICAL PERMITS

All Electrical Installations must be inspected by the Electrical Safety Authority. Separate Inspection Applications must be filed. For more information please call: Electrical Safety Authority: 1-877-ESA-SAFE (1-877-372-7233)

PROTECTION FROM BACKFLOW

Protection from backflow shall be installed for every dwelling unit in accordance with Ontario Building Code, Division B, Section 7.4.6.4 and the referenced CAN/CSA standards. The BWV shall be located within the dwelling unit and be accessible for inspection and maintenance purposes.

NEW FURNACES

All new furnaces must be equipped with an electronically commutated motor and must be designed to CSA-F280-2012.

MAXIMUM TEMPERATURE OF HOT WATER

The maximum temperature of hot water supplied by fittings to fixtures in a residential occupancy shall not exceed 49°C. The maximum temperature does not apply to the hot water supplied to installed dishwashers or clothes washers. (O.B.C. 7.6.5.1)

DECKS, PORCHES & STEPS

The following encroachments are permitted into a yard for decks, porches and steps as specified by the Zoning By-law 8600, as amended.

1. **DECK** means a structure having a floor, open to the sky, and a supporting structure below. It may include perimeter guardrails, fixed seating and vertical visual screening. A patio or terrace is a deck.
 - a. for a deck having a floor height of 0.3 metres or less above *ground* into any required yard, without restriction;
 - b. for a deck or that part thereof having a floor height of greater than 0.3 metres and less than 1.2 metres above the *ground*
 - i. into a *required* front yard, a maximum of 2.5 metres;
 - ii. into a *required* side yard or *required* rear yard;
 - iii. not within 1.2 metres of a any lot line;
 - c. for a deck or part thereof having a floor height of greater than 1.2 metres above the *ground*
 - i. into any front or *required* rear yard, a maximum of 2.5 metres;
 - ii. not within 1.2 metres of a front or rear lot line;
 - iii. a minimum separation from each side lot line, equal to the minimum side yard width(s) *required* within the zoning district in which the lot is located;
2. **PORCH** means a structure, with a floor and a fixed roof directly above, primarily intended to provide sheltered access to a building. It may include perimeter guardrails, a door, seasonal screens or windows
 - a. for a porch,
 - i. into any front or *required* rear yard, a maximum of 2.5 metres;
 - ii. not within 1.2 metres of a front or rear lot line;
 - iii. a minimum separation from each side lot line, equal to the minimum side yard width(s) *required* within the zoning district in which the lot is located;
3. **STEPS**
 - a. for steps having a maximum height of 0.3 metres above the *ground*, into any side yard or *required* front or rear yard, without restriction;
 - b. for steps having a height of greater than 0.3 metres above the *ground*
 - i. into any side yard or *required* front or rear yard, but not within 1.2 metres of a lot line;
 - ii. the steps shall have a maximum total tread area of 5.0 square metres within a *required* yard;-

GROUND means the average elevation of the ground within 3.0 metres of the main building on the lot.

REQUIRED means mandated by the Zoning By-law 8600.

It is the responsibility of the applicant to conform to the requirements of:

2012 Ontario Building Code
Zoning By-law as amended
All other applicable governmental regulations

Water Service 1" Water service required as per O.B.C. 7.6.3.4.

Surface Drainage 9.14.6.1.(1) The building shall be located or the building site graded so that water will not accumulate at or near the building and will not adversely affect adjacent properties.

Lot Grading The owner shall submit to the Building Division, following completion of construction, written verification by a Consulting Engineer or Ontario Land Surveyor of the approved elevation as per the lot grading plan.

Unfinished Basements Separate permit required to Finish Basement

Sump Pits 9.14.5.2 & 9.25.3.3.(16) To be designed to resist removal by children, & pit covers shall be sealed to maintain continuity of the air barrier system.

Drain Water Heat Recovery SB-12 3.1.1.12. The unit shall be installed conforming to CSA B55.2 with an efficiency not less than 42%, according to manufacturer's instructions.

All New Furnaces must be equipped with an electronically commutated motor and must be designed to CSA-F280-2012.

Backflow Protection shall be installed for every dwelling unit in accordance with Ontario Building Code, Division B, Section 7.4.6.4 and the referenced CAN/CSA standards. The B.W.V. shall be located within the dwelling unit and be accessible for inspection and maintenance purposes.

Every floor level containing a bedroom shall have at least 1 window that shall open from inside without tools & provide an unobstructed open portion of a minimum of 3.8 sq. ft. & with no dimension less than 15". Maintains the required opening without the need for additional support. Maximum sill height of 3'-3" above the floor except basement

9.10.17.10 Protection of Foamed Plastics Foamed Plastics to be protected from adjacent space in buildings. Provide Gypsum Board protection or approved alternative

Clothes Dryer & Exhaust Fans Exhaust to Exterior with Non-Combustible Duct

Smoke Alarms 9.10.19. To have battery backup & be interconnected w/ visual signaling component conforming to 18.5.3.(Light, Color and Pulse Characteristics) of NFPA 72, "National Fire Alarm and Signaling Code" Located on each floor w/ CO Detector, hallways outside of bedroom w/ CO Detector, and any room to be used as a bedroom.

Roof Structure Has been approved with engineered truss design, if conventional framing replaces truss design drawings are to be resubmitted for review prior to framing inspection.

All Engineered product shop drawings shall be stamped by a licensed Professional Engineer of Ontario.

Shop drawings will be reviewed for approval by a Building Official.

Products requiring the submission of shop drawings include but are not limited to, roof trusses, floor systems and wood beams.

Prior to the start of framing, email a stamped PDF copy of the shop drawings to buildingdept@citywindsor.ca Email to include Permit Number and Property Address for reference.

Hard copy of shop drawings to be provided to the Building Inspector at the time of framing inspection.

Roof Ventilation O.B.C. 9.19.1.2 Unobstructed vent area not less than 1/300 of insulated ceiling. Where roof slope is slope less than 1 in 6 unobstructed vent area not less than 1/150 of insulated ceiling. Min 25% required openings located at the top or bottom of space. Venting to be uniformly distributed on all sides of building.

EIFS Design and Installation Shall comply with CAN/ULC S716.2 & CAN/ULC S716.3 Inclusive of weather resistive barrier, drainage layer, flashing, etc (no tyvek)

Downspouts To splash to grade in accordance with approved lot grading plan unless hazard exists.

PLUMBING DRAINAGE PLAN ☐

SEWER DRAINAGE PLAN ☐



PERMIT NUMBER

ADDRESS

IMPORTANT

TO OBTAIN AN INSPECTION YOU MUST:

- A. In the space provided, prepare a drainage plan AS INSTALLED
- B. Complete the information on this card
- C. Give 24 hours notice on all inspections
- D. Submit this card to the City Inspector on site prior to Inspection

City Inspector use ONLY

SIGNATURE

DATE

USE INK AND A STRAIGHTEDGE. INCLUDE OUTLINE OF BUILDING AND INDICATE NORTH.

GENERAL NOTES

ALL CONSTRUCTION TO COMPLY WITH THE FOLLOWING GENERAL NOTES AND/OR TO THE CURRENT ONTARIO BUILDING CODE 2012, AND/OR LOCAL GOVERNING CODES IN THE EVENT OF A CONFLICT, THE MORE STRINGENT REQUIREMENTS SHALL APPLY.

GENERAL:

THIS BUILDING HAS BEEN DESIGNED IN ACCORDANCE WITH THE ONTARIO BUILDING CODE 2012. A COPY OF THE CODE BOOK SHOULD BE RETAINED BY THE BUILDER/ GENERAL CONTRACTOR FOR REFERENCE BY THE ON SITE CONSTRUCTION PERSONAL. ALL CONSTRUCTION SHALL CONFORM TO ALL REQUIREMENTS OF THE CURRENT CODE.

MECHANICAL SYSTEM DESIGN IS BY THE MECHANICAL CONTRACTOR. DESIGN AND INSTALLATION SHALL COMPLY WITH PARTS 6, 9 AND ALL OTHER APPLICABLE PARTS OF THE ONTARIO BUILDING CODE 2012.

PLUMBING SYSTEM DESIGN IS BY OTHERS. DESIGN AND INSTALLATION SHALL COMPLY WITH PARTS 7, 9, AND ALL OTHER APPLICABLE PARTS OF THE ONTARIO BUILDING CODE 2012.

THESE NOTES ARE FOR GENERAL REFERENCE. WHERE CONFLICTS EXIST BETWEEN THESE NOTES AND CURRENT CODES THE MORE STRINGENT REQUIREMENTS SHALL PREVAIL.

MATERIALS OR CONSTRUCTION PROCEDURES WHICH ARE PROHIBITED BY LAW OR SHALL CAUSE A HARMFUL EFFECT TO THE NATURAL ENVIRONMENT OR TO THE HEALTH OF ANY PERSON ON THE SITE DURING CONSTRUCTION AND/OR DURING OCCUPANCY SHALL NOT BE USED IN THIS PROJECT.

ALL TRADES SHALL CONFORM WITH ALL APPLICABLE PROVINCIAL, NATIONAL, LOCAL, AND OSHA CODES, RULES AND REGULATIONS. IN CASE OF CONFLICT, THE MOST STRINGENT REQUIREMENTS SHALL APPLY.

DIMENSIONS OF INTERIOR WALLS ON PLANS SHALL BE 3 1/2" (ROUGH STUD DIMENSIONS) UNLESS OTHERWISE INDICATED.

SEE SHEET #3 FOR TYPICAL DETAILS

ELECTRICAL:

ELECTRICAL SYSTEM DESIGN IS BY OTHERS. DESIGN AND INSTALLATION SHALL COMPLY ONTARIO ELECTRICAL SAFETY CODE- CURRENT EDITION

BEDROOM OUTLETS - ALL BRANCH CIRCUITS THAT SUPPLY 125 VOLT, SINGLE PHASE, 15 AND 20 AMPERE OUTLETS INSTALLED IN DWELLING UNIT. BEDROOMS SHALL BE PROTECTED BY AN ARC-FAULT CIRCUIT INTERRUPTER LISTED TO PROVIDE PROTECTION OF THE ENTIRE BRANCH CIRCUIT.

WINDOWS AND GLAZING:

AT FIRST AND SECOND FLOOR A MINIMUM OF ONE(1) WINDOW IN EACH SLEEPING AREA SHALL MEET EGRESS REQUIREMENTS PER SECTION R.3.10. WINDOW CONTRACTOR SHALL PROVIDE EGRESS HARDWARE NECESSARY TO ALLOW WINDOWS TO MEET APPLICABLE EGRESS REQUIREMENTS.

PROVIDE FLASHING AT ALL WINDOW HEAD, JAMB AND SILL CONDITIONS.

FIXED GLASS SIZES SHOWN ARE FOR REFERENCE ONLY. GLAZING CONTRACTOR SHALL FIELD MEASURE ALL ROUGH OPENINGS FOR FIXED GLASS PRIOR TO FABRICATION

OPERATING SASH ARE SHOWN FOR BASIC SIZING ONLY. FINAL SIZE FOR ROUGH OPENING AND GLAZING SIZES SHALL BE PER SELECTED WINDOW MANUFACTURER'S STANDARDS

PROVIDE ALL REQUIRED SAFETY GLASS IN ACCORDANCE WITH ALL APPLICABLE CURRENT BUILDING CODES.

WHERE THE OPENING OF AN OPERABLE WINDOW IS LOCATED MORE THAN 72 INCHES ABOVE THE FINISHED GRADE OR SURFACE BELOW, THE LOWEST PART OF THE CLEAR OPENING OF THE WINDOW SHALL BE A MINIMUM OF 24 INCHES ABOVE THE FINISHED FLOOR OF THE ROOM IN WHICH THE WINDOW IS LOCATED. OPERABLE SECTIONS OF WINDOWS SHALL NOT PERMIT OPENINGS THAT ALLOW PASSAGE OF A 4 INCH DIAMETER SPHERE WHERE SUCH OPENINGS ARE LOCATED WITHIN 24 INCHES OF THE FINISHED FLOOR.

DOORS:

ALL DOORS SHALL BE 6'-8" HIGH AT 8'-0" & 9'-0" CEILINGS.

ALL DOORS SHALL BE 8'-0" HIGH AT 10'-0" CEILINGS UNLESS NOTED OTHERWISE.

ALL OTHER DOOR HEIGHTS SHALL BE COORDINATED W/ OWNER AND /OR GENERAL CONTRACTOR

STAIRS:

ALL STAIRS SHALL CONFORM TO THE REQUIREMENTS OF O.B.C., SECTION 9.8. THE MAXIMUM RISER HEIGHT SHALL BE 7 3/4" AND THE MINIMUM TREAD DEPTH SHALL BE 10". ALL TREADS SHALL HAVE A NOSING AS DESCRIBED IN SECTION 9.8.3

ENCLOSED ACCESSIBLE SPACE UNDER THE STAIR SHALL HAVE WALLS, UNDER STAIR SURFACE AND ANY SOFFITS TO BE PROTECTED ON THE ENCLOSED SIDE WITH 1/2" GYPSUM BOARD. HANDRAILS SHALL HAVE A MINIMUM AND MAXIMUM HEIGHT OF 34" & 38" RESPECTIVELY MEASURED VERTICALLY FROM THE NOSING OF THE STAIR. HANDRAILS) SHALL BE CONTINUOUS THE FULL LENGTH OF THE STAIR, EXCEPT AS ALLOWED BY SECTION 9.8.7

THE HANDGRIP PORTION OF THE HANDRAIL SHALL HAVE A CIRCULAR CROSS SECTION DIMENSION OF 1 1/4" MINIMUM TO 2 5/8" MAXIMUM, OR PROVIDING AN EQUIVALENT GRASPING SURFACE.

GUARD RAIL:

BALUSTERS SHALL BE SPACED SO THAT A SPHERE WITH A DIAMETER OF 4" CANNOT PASS THROUGH ANY OPENING

GUARD RAILS SHALL MEET THE FOLLOWING:

GUARD RAILS AT PORCHES, BALCONIES, OR RAISED FLOOR SURFACE WITH A HEIGHT DIFFERENTIAL OF 30" OR MORE SHALL BE A MINIMUM OF 36" HIGH.

INTERIOR GUARD RAILS SHALL BE 36" HIGH MINIMUM.

SMOKE DETECTOR/ALARMS:

EACH SLEEPING ROOM SHALL BE PROVIDED WITH A MINIMUM OF ONE(1) SMOKE DETECTOR (LOCAL FIRE DEPARTMENT APPROVED AND UNDERWRITERS' LABORATORIES TESTED AND LABELED) AND ONE SMOKE DETECTOR INSTALLED IN COMMON AREA (HALL OR CORRIDOR) ADJACENT TO THE SLEEPING ROOMS (WITHIN 10 FEET OF ALL BEDROOM DOORS. ALSO PROVIDE A MINIMUM OF ONE (1) SMOKE DETECTOR ON EACH FLOOR. THE SMOKE DETECTOR IS TO BE INSTALLED IN ACCORDANCE WITH ALL APPLICABLE CODES. THE DETECTOR SHALL BE WIRED IN SUCH A WAY THAT THE ACTIVATION OF ONE (1) ALARM WILL ACTIVATE ALL THE ALARMS IN THE DWELLING UNIT. ALL SMOKE DETECTORS SHALL BE EQUIPPED WITH A BATTERY BACKUP.

FOAM PLASTICS:

ALL FOAM PLASTICS OR FOAM PLASTIC CORED MATERIAL USED IN BUILDING CONSTRUCTION SHALL HAVE SURFACE BURNING CHARACTERISTICS OR A THERMAL BARRIER AS DESCRIBED IN SECTION O.B.C. 9.10.16 UNLESS NOTED OTHERWISE

FLAME SPREAD AND SMOKE DENSITY:

ALL WALL AND CEILING FINISHES SHALL COMPLY WITH THE REQUIREMENTS OF SECTION 9.10.16

SITE

BRICK OR MASONRY LEDGES SHALL BE A MINIMUM OF 6" ABOVE FINISH GRADE.

FLOOR FRAMING:

WOOD FLOOR FRAMING SHALL COMPLY WITH PART 9, EXCEPT WHERE MORE STRINGENT REQUIREMENTS ARE NOTED ON DRAWINGS.

FRAMING ASSEMBLY FASTENERS SHALL BE INSTALLED PER TABLE 9.23.3.4.

SEE DETAILS AND CODE FOR ALLOWABLE NOTCHING AND BORING OF DIMENSIONAL LUMBER. SECTION 9.23.3 FOR JOIST.

SEE MANUFACTURER'S SPECIFICATIONS FOR ALLOWABLE CUTTING AND BORING OF PRE-ENGINEERED MATERIALS USED IN FLOOR FRAMING.

PROVIDE DRAFTSTOPPING PER SECTION 9.23.7.2

BEARING WALLS: PROVIDE 2X4 LADDER BLOCKING AT 16" O.C. ON 2X4 LEDGER BOARDS BETWEEN

HEADER JOISTS (SEE DRAWINGS FOR SIZE OF MEMBER) UNDER ALL BEARING PARTITIONS PARALLEL TO FLOOR FRAMING DIRECTION.

PROVIDE SOLID BLOCKING UNDER ALL POINT LOAD CONDITIONS CONTINUOUS TO SOLID BEARING AT HEADERS OR FOUNDATION.

PROVIDE SOLID BLOCKING BETWEEN JOIST UNDER ALL BEARING WALLS PERPENDICULAR TO FRAMING DIRECTION.

FLOOR FRAMING SPACED GREATER THAN 16" O.C. AT AREAS RECEIVING HARD TILE OR STONE FINISH SHALL HAVE 2x4 LADDER BLOCKING @ 16" O.C. BETWEEN MEMBERS

ROOFING:

ASPHALT SHINGLES SHALL CONFORM TO ASTM D225. ASPHALT SHINGLES SHALL NOT BE INSTALLED ON ROOF SLOPES BELOW TWO UNITS VERTICAL IN 12 UNITS HORIZONTAL (2:12). DOUBLE LAYER UNDERLAYMENT SHALL BE REQUIRED ON ROOF SLOPES BELOW TWO UNITS VERTICAL IN 12 UNITS HORIZONTAL (4:12). SINGLE LAYER UNDERLAYMENT IS REQUIRED ON ALL OTHER ROOF SLOPES. ASPHALT SHINGLES SHALL BE SECURED TO THE ROOF WITH NOT LESS THAN FOUR FASTENERS PER STRIP SHINGLE, OR NOT LESS THAN TWO FASTENERS PER INDIVIDUAL SHINGLE. SHINGLE HEADLAP SHALL NOT BE LESS THAN 2 INCHES (15 MM).

ROOF PENETRATIONS:

ALL PLUMBING, MECHANICAL VENT STACKS AND FURNACE FLUES SHALL BE OFFSET TO REAR ROOF LINES. FURNACE FLUES SHALL COMPLY WITH CODE FOR MINIMUM SLOPE AND NUMBER OF TURNS ALLOWED FOR OFFSETS.

FLASHING:

ALL TRANSITION FLASHING (ROOF TO WALL) SHALL LAP VERTICAL WALL FACE A MINIMUM OF 8" UNLESS NOTED OTHERWISE.

PROVIDE FLASHING AT ALL EXTERIOR STEEL UNTEL CONDITIONS AND AT CONCEALED STEEL UNTELS CARRYING EXPOSED BRICK.

PROVIDE EAVE FLASHING PER CODE.

WALL FRAMING:

WOOD WALL FRAMING SHALL COMPLY WITH PART 9 EXCEPT WHERE MORE STRINGENT REQUIREMENTS ARE NOTED ON DRAWINGS.

FRAMING ASSEMBLY FASTENERS SHALL BE INSTALLED PER TABLE 9.23.3.4.

SEE DETAILS AND CODE FOR ALLOWABLE NOTCHING AND BORING OF DIMENSIONAL LUMBER. SECTION 9.23.3 FOR WALL STUDS.

SEE MANUFACTURER'S SPECIFICATIONS FOR ALLOWABLE CUTTING AND BORING OF PRE-ENGINEERED MATERIALS USED IN WALL FRAMING.

PROVIDE FIRE BLOCKING PER SECTION 9.10.16

WALL COVERING MATERIALS SHALL COMPLY WITH O.B.C. 9.10.17

EXTERIOR WOOD FRAME WALLS OVER 9'-2" IN HEIGHT SHALL BE A MINIMUM OF 2x6 CONSTRUCTION. ALL STUDS SHALL BE CONTINUOUS FROM FLOOR TO UNDERSIDE OF FLOOR OR ROOF FRAMING ABOVE. SEE FRAMING MATERIALS FOR MINIMUM STUD SIZES AND GRADES.

ALL STRUCTURAL MULLIONS TO HAVE MINIMUM DOUBLE STUD CONSTRUCTION CONTINUOUS FROM FLOOR TO UNDERSIDE OF FLOOR/ROOF FRAMING ABOVE. WINDOW TRANSOM HEADERS SHALL SPAN BETWEEN CONTINUOUS STUDS WITH FLUSH HANGER BRACKETS AS REQUIRED.

PROVIDE CONTINUOUS WALL STUDS FROM FLOOR TO UNDERSIDE OF ROOF FRAMING AT ALL SLOPED CEILING CONDITIONS. (BALLOON CONSTRUCTION)

INTERIOR LOWER LEVEL BEARING WALLS SHALL BE 2X6 FRAMING AT 16" O.C. WHEN CARRYING MORE THAN TWO(2) FLOORS OR ONE (1) FLOOR AND ONE (1) ROOF.

WALL SHEATHING:

PROVIDE STRUCTURAL GRADE OSB WALL SHEATHING FOR LATERAL BRACING OF EXTERIOR WALL LOADS. WHEN NON-STRUCTURAL SHEATHING IS USED PROVIDE LET-IN DIAGONAL BRACING OR OTHER APPROVED TYPE OF BRACING AT ALL EXTERIOR CORNERS OF STRUCTURE.

ROOF-CEILING CONSTRUCTION:

WOOD ROOF FRAMING SHALL COMPLY WITH SECTION 9.23.13 EXCEPT WHERE MORE STRINGENT REQUIREMENTS ARE NOTED ON DRAWINGS.

FRAMING ASSEMBLY FASTENERS SHALL BE INSTALLED PER TABLE 9.23.3.4

ROOF TIE-DOWN REQUIREMENTS SHALL COMPLY WITH CODE

DIMENSIONAL LUMBER:

EXTERIOR - BEARING AND NON-BEARING WALLS:

8'-1 1/8" PLATE HEIGHT OR LESS: 2x6 SPRUCE-PINE-FIR #1/2 KD OR BETTER

9'-1 1/8" PLATE HEIGHT OR LESS: 2x6 SPRUCE-PINE-FIR #1/2 KD OR BETTER

16'-1 1/8" PLATE HEIGHT OR LESS: 2x8 HEM-FIR #1/2 KD OR BETTER

18'-8" PLATE HEIGHT OR LESS: 2x8 DOUGLAS FIR LARCH #1/2 KD OR BETTER

WALLS INTERIOR - BEARING WALLS : SPRUCE-PINE-FIR #2 KD OR BETTER NON-BEARING WALLS : SPRUCE-PINE-FIR, KILN DRIED, STUD GRADE OR BETTER.

HEADER: HEM-FIR #2 KD OR BETTER FIBER BENDING STRESS=850 P.S (SINGLE MEMBER) ELASTICITY MODULUS=1,300,000 P.S.I.

JOIST AND RAFTERS: HEM-FIR #2 KD OR BETTER: FIBER BENDING STRESS= 1,075 P.S.I. (REPETITIVE MEMBER) ELASTICITY MODULUS=1,300,000 P.S.I.

WALL PLATES, NON-STRUCTURAL BLOCKING: SPRUCE-PINE-FIR, KILN DRIED, UTILITY GRADE OR BETTER.

PERIMETER SILL PLATES: PRESURE TREATMENT AWP, LP-2, KILN DRIED TO 19% MOISTURE CONTENT. SET PERIMETER SILL PLATES ON SILL SEALER.

FURRING: SPRUCE-PINE-FIR, KILN DRIED, NO.3 OR BETTER.

ALL LUMBER GRADES AND STANDARDS BASED ON 'CSA-STANDARD' DESIGN SPECIFICATIONS

(2) 2x8 HEADERS TO BEAR ON (2) TWO JACK STUDS UNLESS NOTED OTHERWISE

(2) 2x10 HEADERS TO BEAR ON (2) TWO JACK STUDS UNLESS NOTED OTHERWISE

ALL PRE-ENGINEERED HEADERS TO BEAR ON THE REQUIRED NUMBER OF STUDS TO MATCH WIDTH OF HEADER MATERIAL AT PERPENDICULAR WALLS AND ON A MINIMUM OF TWO (2) JACK STUDS AT PARALLEL WALL CONDITION UNLESS NOTED OTHERWISE.

ALL PRE-ENGINEERED LUMBER HEADERS SHALL BE BUILT-UP FROM THE NUMBER OF HEADERS INDICATED ON DRAWINGS. ALL MEMBERS SHALL BE SECURED WITH NAILS OR BOLTS AS SPECIFIED BY THE MANUFACTURER FOR SIZES INDICATED.

ALL GIRDER TRUSSES TO BEAR ON (2) TWO STUDS MINIMUM OR AS REQUIRED TO MATCH NUMBER OF TRUSS PLYS UNLESS NOTED OTHERWISE ON THE DRAWINGS OR ON TRUSS DESIGN DRAWINGS.

ALL STRUCTURAL HANGERS TO BE 'SIMPSON' OR APPROVED EQUAL.

CARPENTER CONTRACTOR TO INSTALL NAIL SIZES & NUMBER REQ'D AS SPECIFIED FOR EACH TYPE OF HANGER

INSULATION:

INSULATION NOTE:

PROVIDE INSULATION AS REQUIRED TO MEET ONTARIO BUILDING CODE- 2012 CODE.

PROVIDE RIGID INSULATION AT ALL EXPOSED PERIMETER SLAB ON GRADE CONDITIONS AS REQUIRED TO MEET CURRENT ENERGY CODE REQUIREMENTS.

PROVIDE INSULATION AT ALL BOND CONDITIONS-SEE INSULATION NOTE. PROVIDE INSULATION AROUND ALL SKYLIGHT SHAFTS-SEE INSULATION NOTE.

ALL INSULATION SHALL HAVE A FLAME SPREAD INDEX OF 25 OR LESS AND A SMOKE-DEVELOPED INDEX NOT TO EXCEED 450.

INSULATION SHALL BE INSTALLED IN SUCH A MANNER AS TO ALLOW FREE AIR FLOW FROM THE SOFFIT TO THE ROOF/ATTIC SPACE.

VENTILATION OF CONCEALED ROOF SPACES SHALL BE MAINTAINED.

TRUSS NOTES:

	RTC	DENOTES RAISED HEEL HEIGHT OF TRUSS (REFERENCE TO ROOF TRUSS)
	TCB	DENOTES TOP CHORD BEARING OF TRUSS (REFERENCE TO ROOF OR FLOOR TRUSS)
	BCB	DENOTES BOTTOM CHORD BEARING OF TRUSS (REFERENCE TO FLOOR TRUSS ONLY)
	FG	DENOTES FLOOR GIRDER TRUSS
	GT	DENOTES GIRDER TRUSS

FIRERATED TRUSS SYSTEMS:

TRUSS DESIGNER IS TO REVIEW ALL FIRERATED ASSEMBLIES INDICATED ON THE DRAWINGS. THE TRUSS DESIGN DRAWINGS SHALL PROVIDE CHORD MEMBERS AND INDICATE WHERE ANY BRIDGING, BLOCKING OR STIFFENERS ARE TO BE INSTALLED AS REQUIRED BY THE TESTED ASSEMBLY.

ALL METAL TRUSS PLATES USED IN THE FABRICATION OF THE TRUSSES SHALL BE AS SPECIFIED IN THE TESTED ASSEMBLY, ANY LOAD REDUCTION IN CAPACITY OF PLATES SHALL BE COMPLIED WITH.

WHERE PROPRIETARY PLATES, MATERIALS OR PROCESSES ARE LISTED AS A PART OF THE FIRERATED DESIGN THE TRUSS FABRICATOR SHALL OBTAIN MATERIALS OR CERTIFICATION OF COMPLIANCE WITH THE SYSTEM FROM THE PROPRIETARY HOLDER.

BUILDING DESIGNER'S ASSUMPTION:

1. ROOF TRUSSES TO CLEARSPAN UNLESS NOTED OTHERWISE.

2. FLOOR TRUSSES TO CLEARSPAN BETWEEN BEARING WALLS AS SHOWN. ONE PIECE MULTIPLE SPANS TRUSSES ARE ALLOWED WITH APPROVAL OF BUILDING DESIGNER. DETAILS INDICATING AIR SPACES FOR SOUND CONTROL SHALL NOT BE VIOLATED WITH TRUSS MEMBERS.

3. ROOF AND FLOOR TRUSSES IN FIBERATED SYSTEMS SHALL MEET OR BE LESS THAN THE MAXIMUM SPACING AND MEET OR EXCEED MINIMUM DEPTH REQUIREMENTS AS LISTED IN THE TEST REPORTS. TRUSS PLATES SHALL MEET CRITERIA LISTED IN TEST REPORTS.

4. WIND LOADS ARE BEING TRANSFERRED THROUGH EXTERIOR WALLS WITH TRUSS CONNECTION POINTS AND THROUGH INTERIOR SHEAR WALLS WHERE INDICATED ON PLANS. WHERE ROOF SHEATHING IS INDICATED TO TRANSFER TO INTERIOR SHEAR WALL TRUSS CHORDS SHALL BE SIZED FOR NAILING PATTERNS AS INDICATED TO DIAPHRAGM TRUSS AND TRANSFER LOADS.

5. GABLE-END TRUSSES: ALL GABLE TRUSSES TO BE SHEATHED WITH 7/16" OSB MATERIAL. VERTICAL REINFORCING OF WEB MEMBERS TO BE DESIGNED BY TRUSS DESIGNER TO PREVENT WITHDRAWAL LOADS OCCURRING IN TRUSS PLATES. LATERAL BRACING OF TOP AND BOTTOM TRUSS CHORDS TO BE TRANSFERRED THROUGH ROOF AND CEILING SHEATHINGS. DIAGONAL BRACING IS DESIGNED BY TRUSS DESIGNER AND SHALL REMAIN IN PLACE TO PREVENT RACKING OF TRUSS UNDER LATERAL LOADING.

6. BUILDING DESIGNER IS NOT RESPONSIBLE FOR TRUSS CHORD MEMBERS SUBJECT TO DESIGN DEFICIENCIES. REINFORCING IF REQUIRED IS THE RESPONSIBILITY OF TRUSS DESIGNER.

7. ANY DIFFERENCES BETWEEN CODE REQUIREMENTS AND TRUSS INDUSTRIES STANDARDS THE MORE STRINGENT SHALL APPLY.

8. IT IS IMPORTANT FOR THE TRUSS DESIGNER/FABRICATOR TO TAKE GREAT CARE IN THE BANDING, SHIPPING, AND DELIVERY PROCESS TO INSURE THE TRUSSES ARE NOT DAMAGED. SEE HIB-91, PROVIDED BY TRUSS DESIGNER/FABRICATOR, FOR PROPER STORAGE METHODS FOR TRUSSES PRIOR TO DELIVERY AND ERECTION.

FLOOR & ROOF TRUSSES:

THE TRUSS DESIGNER IS TO PROVIDE A DESIGN FOR AN ENTIRE ROOF OR FLOOR SYSTEM, AND NOT FOR INDIVIDUAL COMPONENTS. THE TRUSS DESIGNER MUST ASCERTAIN THAT THE LOADS UTILIZED MEET OR EXCEED THE LOAD VALUES REQUIRED BY ONTARIO BUILDING CODE.

TRUSS MANUFACTURER SHALL BE RESPONSIBLE FOR ALL TRUSS DESIGNS INCLUDING GIRDERS, HANGERS, BEARING SEATS, AND ANCHORS FOR TRUSSES.

TRUSS FRAMING SHOWN ON PLANS IS FOR GENERAL REFERENCE AND TO INDICATE BEARING LOCATIONS. TRUSS MANUFACTURER SHALL NOTIFY ARCHITECT IF ADDITIONAL BEARING POINTS AND/OR WALLS ARE NEEDED PRIOR TO FABRICATION AND ERECTION.

ALL ROOF TRUSSING SHALL BE BRACED PER THE TRUSS DESIGNER/FABRICATOR'S REQUIREMENTS, BCSI 1-03 AND AS REQUIRED ON DRAWINGS.

RUSS MEMBERS SHALL NOT BE CUT, NOTCHED, DRILLED, SPLICED OR OTHERWISE ALTERED IN ANY WAY WITHOUT THE APPROVAL OF THE LICENSED TRUSS DESIGNER OR STRUCTURAL ENGINEER.

TRUSS DESIGN DRAWINGS, PREPARED IN COMPLIANCE WITH SECTION R-802.1.1, SHALL BE PROVIDED TO THE BUILDING OFFICIAL AND APPROVED PRIOR TO INSTALLATION. TRUSS DESIGN DRAWING SHALL BE PROVIDED WITH THE SHIPMENT OF TRUSSES DELIVERED TO THE JOBSITE. TRUSS DESIGN DRAWINGS SHALL INCLUDE, AT A MINIMUM, THE FOLLOWING INFORMATION SPECIFIED BELOW:

- SLOPE OR DEPTH, SPAN, AND SPACING.
- LOCATION OF ALL JOINTS.
- REQUIRED BEARING WIDTHS.
- DESIGN LOADS AS APPLICABLE.
 - TOP CHORD LIVE LOAD (INCLUDING NORMAL SNOW LOADS & DRIFTING LOADS)
 - TOP CHORD DEAD LOAD.
 - BOTTOM CHORD LIVE LOAD.
 - BOTTOM CHORD DEAD LOAD.
 - CONCENTRATED LOADS AND THEIR POINTS OF APPLICATION.
 - CONTROLLING WIND AND/OR EARTHQUAKE LOADS.

5. ADJUSTMENTS TO LUMBER AND JOINT CONNECTOR DESIGN VALUES FOR CONDITIONS OF USE.

6. EACH REACTION FORCE AND DIRECTION.

7. JOINT CONNECTOR TYPE AND DESCRIPTION (E.G., SIZE, THICKNESS OR GAUGE); AND THE DIMENSIONED LOCATION OF EACH JOINT CONNECTOR EXCEPT WHERE SYMMETRICALLY LOCATED RELATIVE TO THE JOINT INTERFACE.

8. LUMBER SIZE, SPECIES AND GRADE FOR EACH MEMBER.

9. CONNECTION REQUIREMENTS FOR:

- TRUSS-TO-TRUSS GIRDER
- TRUSS PLY-TO-PLY.
- FIELD SPLICES.
- CALCULATED DEFLECTION RATIO AND OR MAXIMUM DESCRIPTION FOR LIVE AND TOTAL LOAD.

11. MAXIMUM AXIAL COMPRESSION FORCES IN THE TRUSS MEMBERS AND ANY REINFORCING REQUIRED FOR OVERSTRESSED MEMBERS. CONNECTIONS AND ANCHORAGE OF THE PERMANENT CONTINUOUS LATERAL BRACING. FORCES SHALL BE SHOWN ON THE TRUSS DRAWING OR ON SUPPLEMENTAL DOCUMENTS.

12. REQUIRED PERMANENT TRUSS MEMBER BRACING LOCATION.

TRUSS FABRICATOR/CONTRACTOR TO PROVIDE ALL HANGERS W/ MODEL No., CLEARLY STAMPED & LAYOUT DRAWINGS CLEARLY INDICATING LOCATION OF VARIOUS HANGERS REQUIRED.

TRUSS FABRICATOR/CONTRACTOR TO PROVIDE TRUSS LAYOUT DRAWINGS SHOWING THE FOLLOWING (IN ADDITION TO ANY CODE REQUIREMENTS):

- TRUSS TYPE INDICATORS CLEARLY MARKED.
- END LOAD REACTIONS FOR ALL GIRDER TRUSSES, INCLUDING ANY UPLIFT LOADS.

ROOF TRUSSES:

- TRUSS SPACING SHALL BE DETERMINED BY THE TRUSS DESIGNER AND NOT EXCEED 24" O.C. AS REQUIRED FOR ROOF SHEATHING.
- PIGGY-BACK TRUSSES SHALL BEAR ON PERPENDICULAR BRACING INSTALLED ON TOP CHORD OF LOWER MAIN TRUSS. MAIN TRUSS SHALL BE DESIGNED AS REQUIRED. WHERE TRUSS WEBS EXCEED ALLOWABLE AXIAL LOADS TRUSS DESIGNER SHALL SPECIFY REQUIRED STIFFENERS OR BRACING.
- THE 1/2" STRUCTURAL ROOF SHEATHING SHALL BE INSTALLED AND IS INTENDED TO BECOME THE PERMANENT BRACING FOR THE TRUSS TOP CHORD
- THE 5/8" MIN. GYPSUM BOARD SHALL BE INSTALLED AND IS INTENDED TO BECOME THE PERMANENT BRACING FOR THE TRUSS BOTTOM CHORD
- TRUSS DESIGNER SHALL DESIGN ALL TRUSSES FOR LOADS AND SPANS AS REQUIRED TO COMPLY WITH THE INTENT OF THE DRAWINGS. IT SHALL BE THE RESPONSIBILITY OF THE TRUSS DESIGNER TO SIZE WEB MEMBERS TO BE STRUCTURALLY ADEQUATE FOR LOADS IMPOSED. OVER STRESSED MEMBERS SHALL HAVE NECESSARY REINFORCEMENT DESIGNED BY THE TRUSS DESIGNER.
- GABLE END TRUSSES SHALL BE DESIGNED TO RECEIVE WALL SHEATHING. BRACING FOR LATERAL LOADS SHALL DESIGNED BY TRUSS DESIGNER TO ELIMINATE OUT-OF-PLANE LOADS ON TRUSS PLATES.
- GABLE END TRUSSES SHALL BE DESIGNED TO MATCH THE CEILING PROFILE OF ROOM ADJACENT TO END WALL.
- TRUSS DESIGNER SHALL PROVIDE DESIGNS FOR PERMANENT LATERAL BRACING TO FUNCTION WITH THE PROPOSED TRUSS SYSTEM. TRUSS CHORDS RECEIVING LOADS FROM BRACING SHALL BE SIZED ACCORDINGLY.
- TRUSS DESIGNER SHALL PROVIDE DESIGNS FOR LATERAL BRACING TO RUN CONTINUOUS ALONG TRUSS CHORDS WHERE PRACTICAL TO MAINTAIN LONGEST POSSIBLE BRACING LINE THROUGH TRUSS SYSTEM WHERE BRACING LINE CANNOT BE MAINTAINED TRUSS DESIGNER IS TO PROVIDE DETAILS FOR TRANSFER OF LOADS IN BRACE LINE OR TERMINATION USING DIAGONAL BRACES.
- TRUSS DESIGNER SHALL DESIGN DIAGONAL BRACING AT TERMINATION POINTS OF PERMANENT LATERAL BRACING. TEMPORARY AND PERMANENT DIAGONAL BRACING SHALL BE INDICATED ON THE ROOF FRAMING LAYOUT PLANS, INCLUDING MEMBER SIZES.
- TRUSS DESIGNER SHALL PROVIDE WOOD GRADE QUALITY OF ALL BRACING MEMBERS TO ELIMINATE SPLITTING AND CRACKING DURING INSTALLATION BY CARPENTER.
- THE TRUSS DESIGNER UNDERSTANDS THAT THE BUILDING DESIGNER HAS NO KNOWLEDGE OF THE CRITERIA AND ASSUMPTIONS MADE IN THE DESIGN OF THE TRUSSES FOR THIS BUILDING. THEREFORE TRUSS CHORD MEMBERS AND PLATES SIZING TO ACCOMMODATE THE STATED REQUIRED BRACING REMAINS THE RESPONSIBILITY OF THE TRUSS DESIGNER.

NOTES TO TRUSS DESIGNER, TRUSS FABRICATOR, AND CARPENTER/TRUSS INSTALLER:

1. LATERAL BRACING SHALL BE AS SHOWN IN DIAGRAMS OF "HIB-91" PUBLICATION PROVIDED BY TRUSS FABRICATOR.

- SEE TABLE, HIB-91 SUMMARY SHEET FRAME 3, "PITCHED TRUSS TOP CHORD TEMPORARY BRACING" FOR MAXIMUM SPACING.
SEE TABLE, HIB-91 SUMMARY SHEET FRAME 4, "BOTTOM CHORD TEMPORARY BRACING" FOR MAXIMUM SPACING.
 - BRACING MEMBER SHALL BE A MINIMUM OF 10'-0" LONG W/ A LAP OF ONE TRUSS BAY EACH END.
 - TEMPORARY BRACING SIZES FOR ROOF TRUSSES SHALL BE SPECIFIED BY TRUSS DESIGNER FOR INSTALLATION TO INTERIOR CHORD FACE OF TRUSSES.
 - SEE TABLE, HIB-91 SUMMARY SHEET FRAME 3, "PITCHED TRUSS TOP CHORD TEMPORARY BRACING" FOR MAXIMUM SPACING OF DIAGONAL BRACING.
 - SEE TABLE, HIB-91 SUMMARY SHEET FRAME 4, "BOTTOM TRUSS CHORD TEMPORARY BRACING" FOR MAXIMUM SPACING OF DIAGONAL BRACING.
 - SEE TABLE, HIB-91 SUMMARY SHEET FRAME 4 FOR DIAGONAL BRACING AT LATERAL BRACE LINES AND MAXIMUM SPACING.
2. TRUSS DESIGNER SHALL PROVIDE COPIES OF ALL FIELD CORRECTIONS AND/OR REPAIR DIAGRAMS AND WRITTEN DIRECTIONS MADE DURING FIELD CONSTRUCTION.

PRE-ENGINEERED TRUSSES:

APPLICANT WHO IS SUBMITTING PLANS FOR PERMITS SHALL PROVIDE AT TIME OF SUBMISSION ONE OF THE FOLLOWING:

- TRUSS DESIGN DATA SHEET
 - COMPLETE TRUSS DESIGN DRAWINGS
- BUILDER OR OWNER SHALL SUBMIT COMPLETE TRUSS DESIGN DRAWINGS PRIOR TO INSTALLATION OF TRUSSES AS REQUIRED BY CODE.

IT IS IMPORTANT FOR THE TRUSS DESIGNER/FABRICATOR TO TAKE GREAT CARE IN THE BANDING, SHIPPING, AND DELIVERY PROCESS TO INSURE THE TRUSSES ARE NOT DAMAGED. SEE HIB-91, PROVIDED BY TRUSS DESIGNER/FABRICATOR, FOR PROPER STORAGE METHODS FOR TRUSSES PRIOR TO DELIVERY AND ERECTION.

LOADING CONDITIONS:

	LIVE LOAD	DEAD LOAD	TOTAL
FLOOR HABITABLE	40 PSF	15 PSF	55 PSF
FLOOR WITH FINISH MAT. ON GROUT BED	40 PSF	25 PSF	65 PSF
ROOF PITCHED	23.5 PSF 10 PSF	T.C. B.C.	17 PSF 40.5 PSF
FLAT ROOF OVER 30' DEEP	23.5 PSF 10 PSF	T.C. B.C.	17 PSF 40.5 PSF

NOTE: ATTICS ARE DESIGNED AS NON-STORAGE, UNLESS NOTED OTHERWISE.
T.C. - TOP CHORD OF TRUSS
B.C. - BOTTOM CHORD OF TRUSS
TRUSS DESIGNER TO INCLUDE A 10 PSF NON-CONCURRENT LOAD IN TRUSS DESIGN

DEFLECTION CRITERIA:

- LIVE LOAD = L/480
- DEAD LOAD = L/360
- USE LIVE LOAD OF L/600 @ TILED AREAS W/ IN FLOOR HEAT

LOT CALCULATIONS

ZONING	PROVD
LOT AREA	3,465.52 ft²
BUILDING A FOOTPRINT	672.0 ft²
BUILDING B FOOTPRINT	429.5 ft²
TOTAL BUILDING FOOTPRINT	1,101.5 ft²
LOT COVERAGE	31.78 %
FRONT YARD	19'-1 (5.8m)
REAR YARD	3'-9" (1.2m)
MIN SDE YARD	3'-9" (1.2m)

STRUCTURAL NOTES:

(2) 2x8 HEADERS TO BEAR ON (1) ONE JACK STUD UNLESS NOTED OTHERWISE.

(2) 2x10 & LARGER HEADERS TO BEAR ON (2) TWO JACK STUDS UNLESS NOTED OTHERWISE.

ALL PRE-ENGINEERED HEADERS TO BEAR ON REQUIRED NUMBER OF STUDS TO MATCH WIDTH OF HEADER MATERIAL OR ON (2) TWO JACK STUDS AT PARALLEL WALL CONDITION UNLESS NOTED OTHERWISE.

ALL PRE-ENGINEERED LUMBER HEADERS SHALL BE BUILT UP FROM THE NUMBER OF HEADERS INDICATED ON DRAWINGS. ALL MEMBERS SHALL BE SECURED WITH NAILS OR BOLTS AS SPECIFIED BY THE MANUFACTURER FOR SIZES INDICATED.

ALL GIRDER TRUSSES TO BEAR ON (2) TWO STUDS MINIMUM OR AS REQUIRED TO MATCH NUMBER OF TRUSS PLYS UNLESS NOTED OTHERWISE.

ALL POINT LOADED HEADERS TO BE LVL.

TRUSS FABRICATOR/CONTRACTOR TO PROVIDE ALL HANGERS W/ MODEL NO. CLEARLY STAMPED & LAYOUT DRAWINGS CLEARLY INDICATING LOCATION OF VARIOUS HANGERS REQUIRED.

CARPENTER CONTRACTOR TO INSTALL NAIL SIZES & NUMBER REQ'D. AS SPECIFIED FOR EACH TYPE OF HANGER.

NON LOADBEARING WALLS PARALLEL TO THE FLOOR JOISTS SHALL BE SUPPORTED BY JOISTS BENEATH THE WALL OR ON BLOCKING BETWEEN THE JOISTS AT 48" O.C.

GENERAL NOTES:

ALL INTERIOR PARTITION DIMENSIONS ARE 3/16" UNLESS NOTED OTHERWISE.

BEFORE CONSTRUCTION OBTAIN WINDOW & DOOR ROUGH OPENING SIZES FROM WINDOW/DOOR SUPPLIER.

SEE SHEET #1 FOR ADDITIONAL NOTES & DETAILS REGARDING DRAWINGS & CONSTRUCTION.

DO NOT SCALE THESE DRAWINGS. USE CALCULATED DIMENSIONS. IF VARIATIONS OCCUR CONTACT ARCHITECT

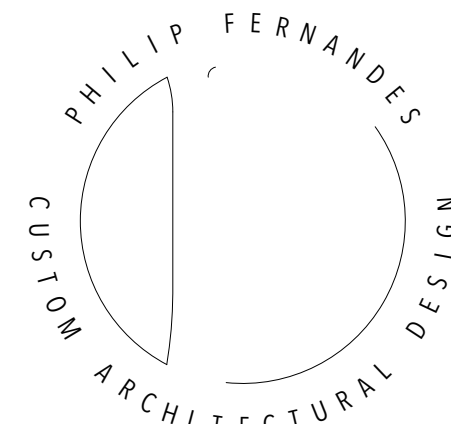
FOR CLARIFICATION:

ALL STAIRS, RAMPS, HANDRAILS, AND GUARDS SHALL CONFORM TO 9.8 OBC.

INTERCONNECTED SMOKE ALARM REQUIRED ON ALL FLOOR LEVELS, CARBON MONOXIDE DETECTORS REQUIRED AT BEDROOM AREAS AS PER OBC 9.10.19.3.

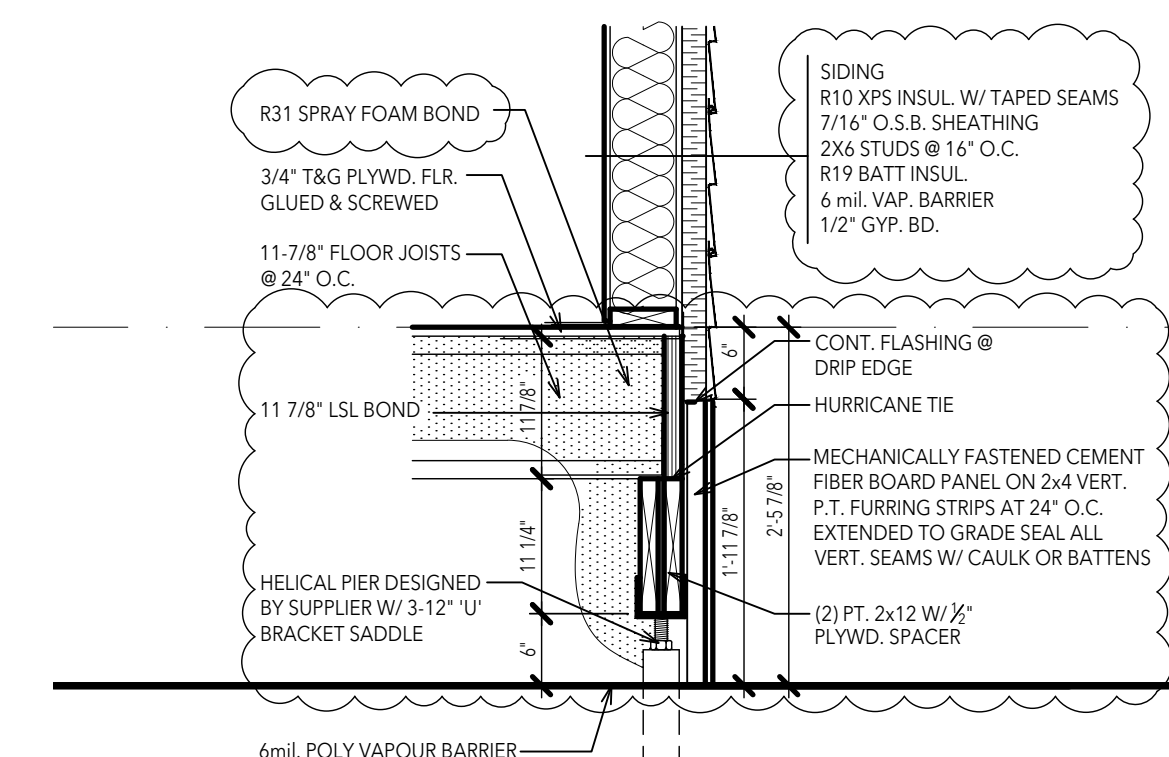
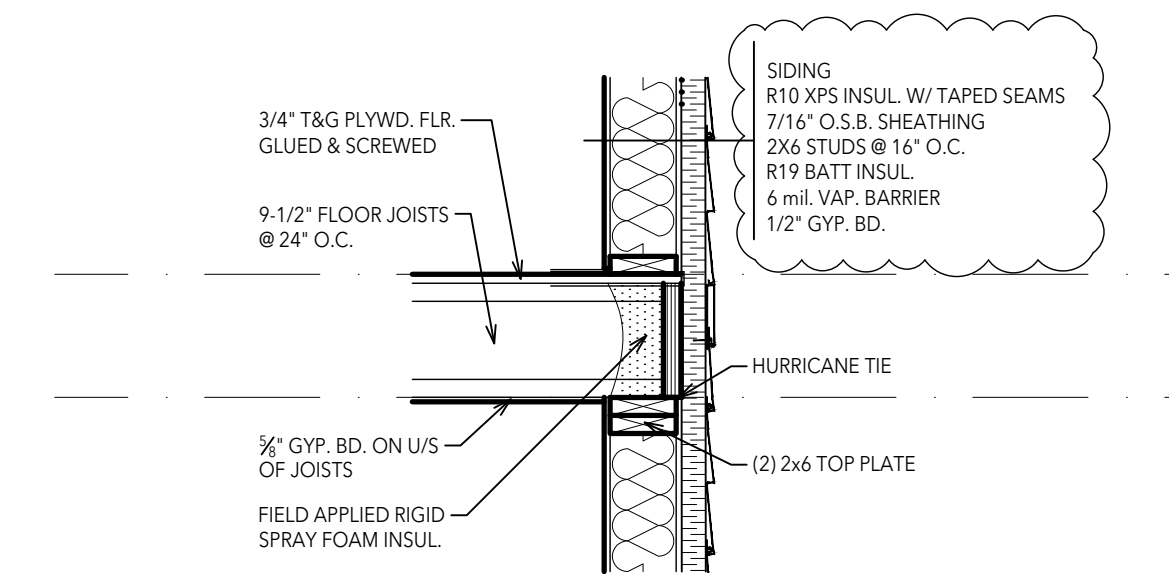
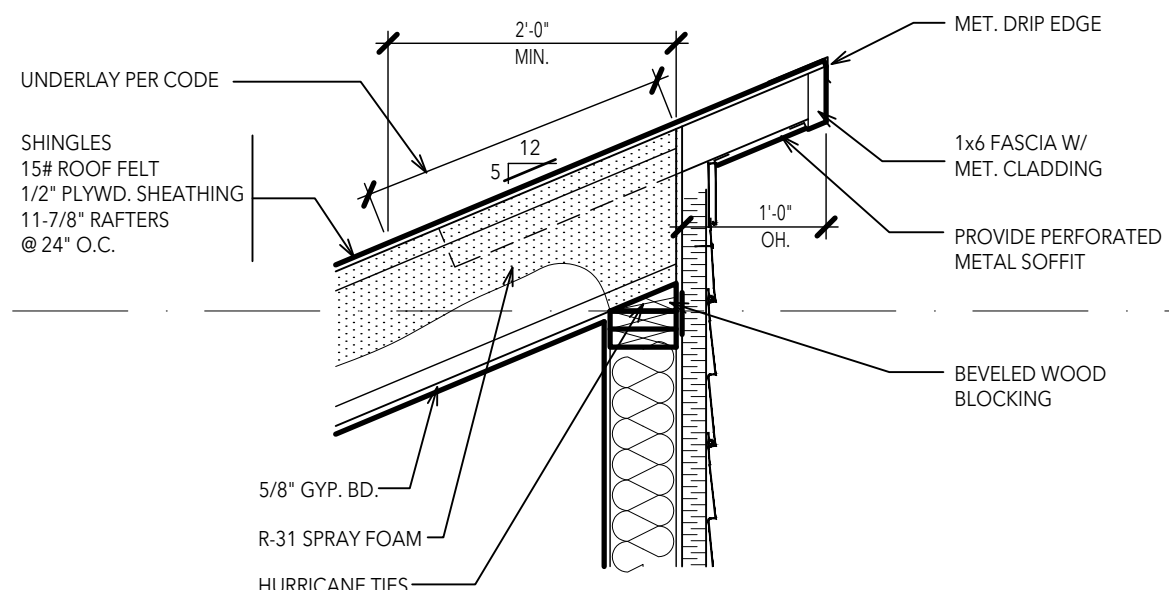
SQUARE FOOTAGE

FIRST FLOOR - SFD	429.5 SQ FT
SECOND FLOOR - ADU	206.4 SQ FT
TOTAL	635.9 SQ FT

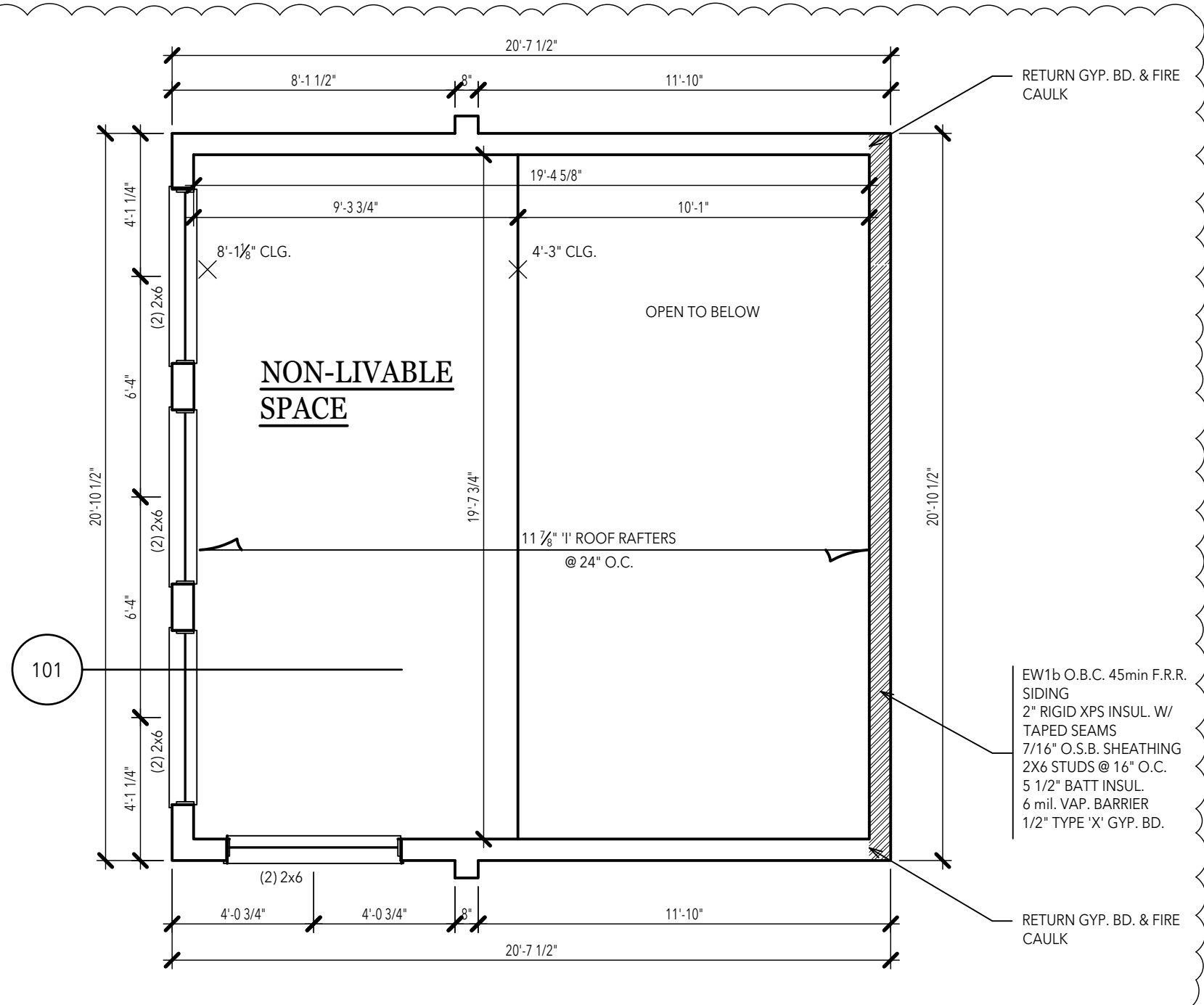


B. COMPLIANCE OPTION (INDICATE THE BUILDING CODE COMPLIANCE PACKAGE BEING EMPLOYED IN THE HOUSE DESIGN)				
SB-12 PRESCRIPTIVE (INPUT DESIGN PACKAGE) PACKAGE: A1 TABLE: 3.1.1.2.A				
C. PROJECT DESIGN CONDITIONS				
CLIMATE ZONE (SB-1): ☒ ZONE 1 (< 5000 DEGREE DAYS) ☐ ZONE 2 (≥ 5000 DEGREE DAYS)	HEATING EQUIPMENT EFFICIENCY ☒ ≥ 92% AFUE ☐ ≥ 84% < 92% AFUE	SPACE HEATING FUEL SOURCE ☐ GAS ☐ PROPANE ☐ SOLID FUEL ☐ OIL ☒ ELECTRIC ☐ EARTH ENERGY		
RATIO OF WINDOWS, SKYLIGHTS & GLASS (W, S & G) TO WALL AREA				
AREA OF WALLS =	1,074.9 ft ²	W, S & G % =	7.3%	
AREA OF W, S & G =	79 ft ²	UTILIZE WINDOW AVERAGING:	☐ YES ☒ NO	
D. BUILDING SPECIFICATIONS (PROVIDES VALUES AND RATINGS OF THE ENERGY EFFICIENCY COMPONENTS PROPOSED)				
BUILDING COMPONENT	MINIMUM RS/R VALUES OR MAXIMUM U-VALUE ¹		BUILDING COMPONENT	EFFICIENCY RATINGS
THERMAL INSULATION	NOMINAL	EFFECTIVE	WINDOWS & DOORS (PROVIDE U-VALUE ¹ OR EER RATING)	
CEILING WITH ATTIC SPACE	R60 + HH	R59.90	WINDOWS/SLIDING GLASS DOORS	U0.25 E.R. 29
CEILING WITHOUT ATTIC SPACE	R31	R27.65	SKYLIGHTS/GLAZED ROOFS	0.49U
EXPOSED FLOOR	R31	R29.8	MECHANICALS	
WALLS ABOVE GRADE	R19 + 10 ci	R25.32	HEATING EQUIP. (AFUE)	-%
BASEMENT WALLS	R20 + 8 ci	R22.71	HRV EFFICIENCY (SRE% AT 0° C)	81%
SLAB (ALL > 600mm BELOW GRADE)	7.5	8.63	DHW HEATER (EF)	0.8 EF
SLAB (EDGE ONLY ≤ 600mm BELOW GRADE)	R10	R10	DWHR (CSA B55.1 (MIN. 42% EFFICIENCY))	42% # SHWR 2
SLAB (ALL ≤ 600mm BELOW GRADE, OR HEATED)	R10	R11.13	COMBINED HEATING SYSTEM	-

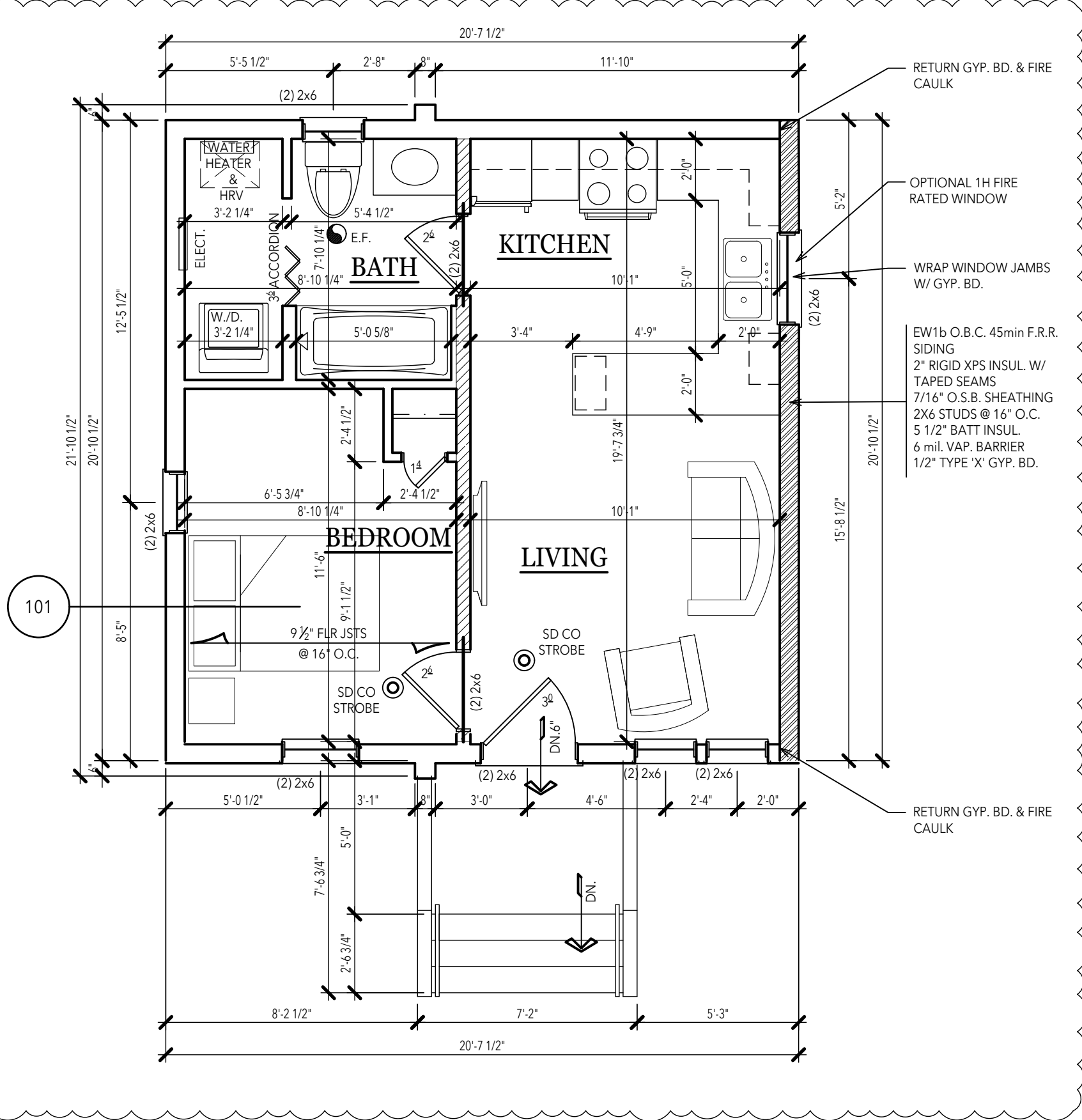
NOTES:
1. U VALUE TO BE PROVIDED IN EITHER W/(m²K) OR Btu/(HrFt²F) BUT NOT BOTH



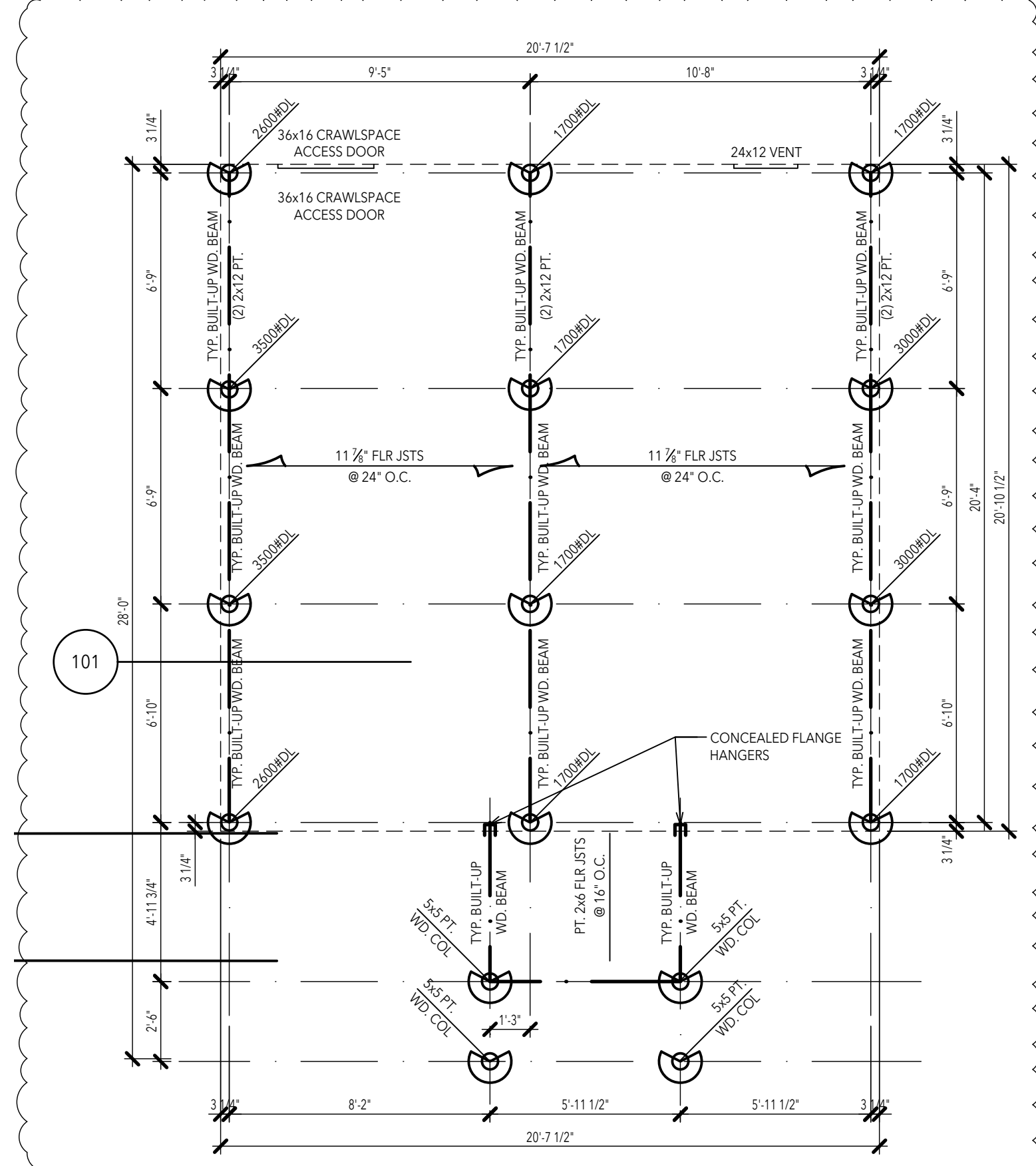
101 TYPICAL BEAM DETAIL
SCALE: 3/4"=1'-0"



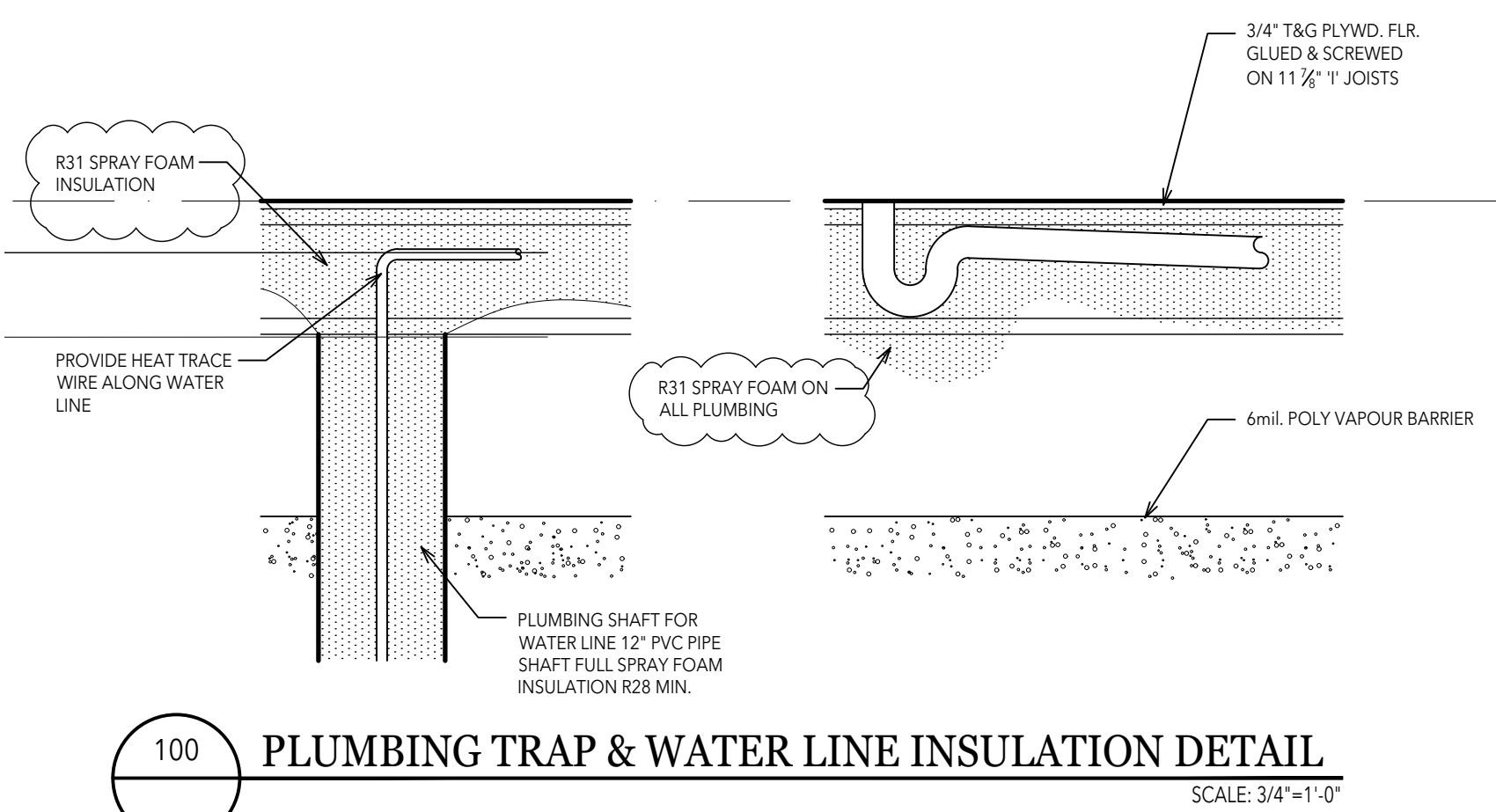
SECOND FLOOR LOFT PLAN
SCALE 1/4"=1'-0"



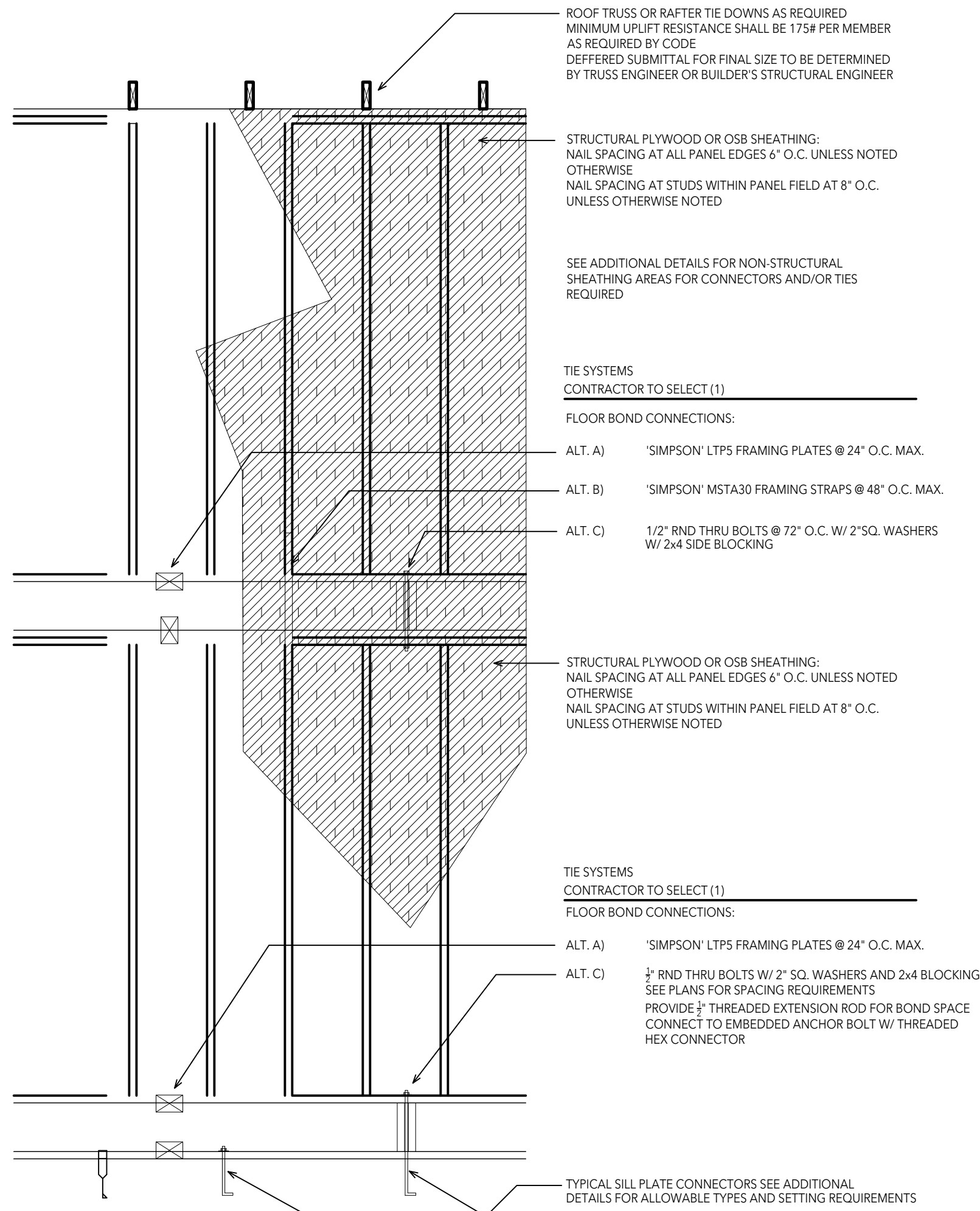
FIRST FLOOR PLAN
SCALE 1/4"=1'-0"



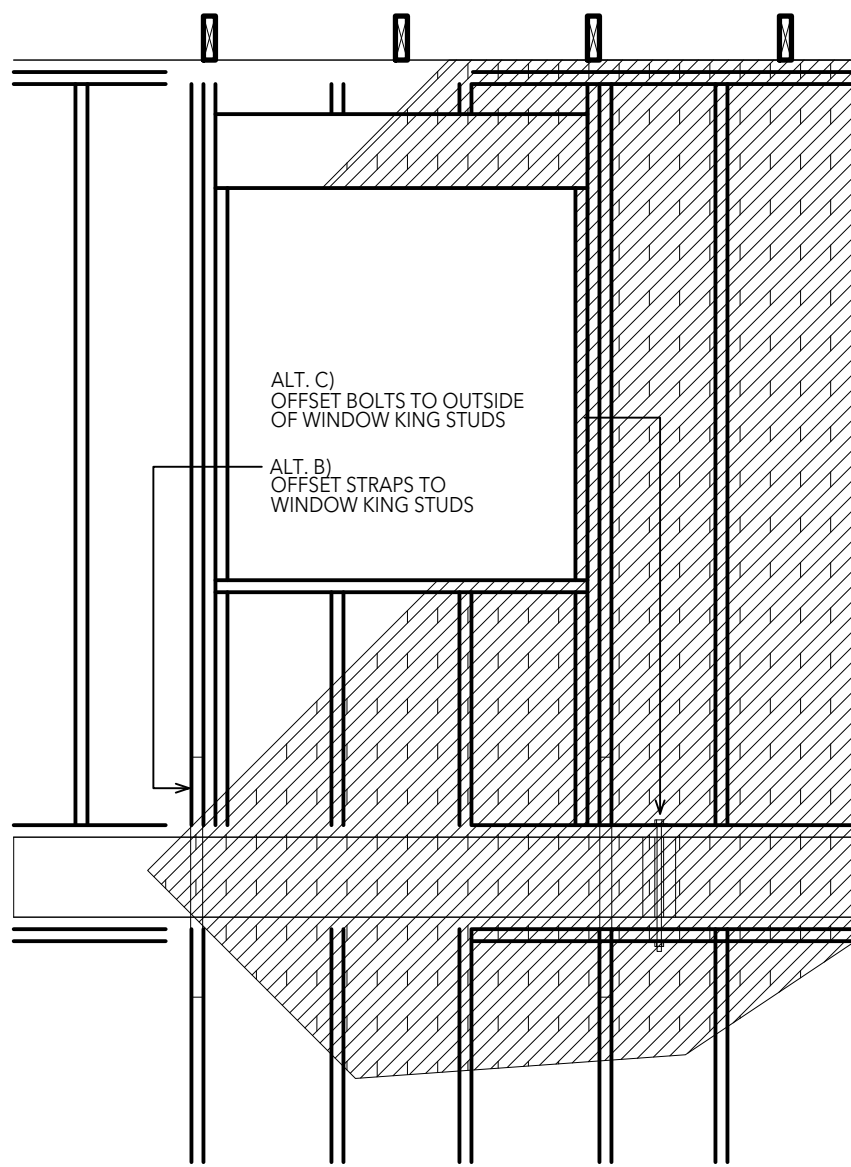
FOUNDATION PLAN
SCALE 1/4"=1'-0"



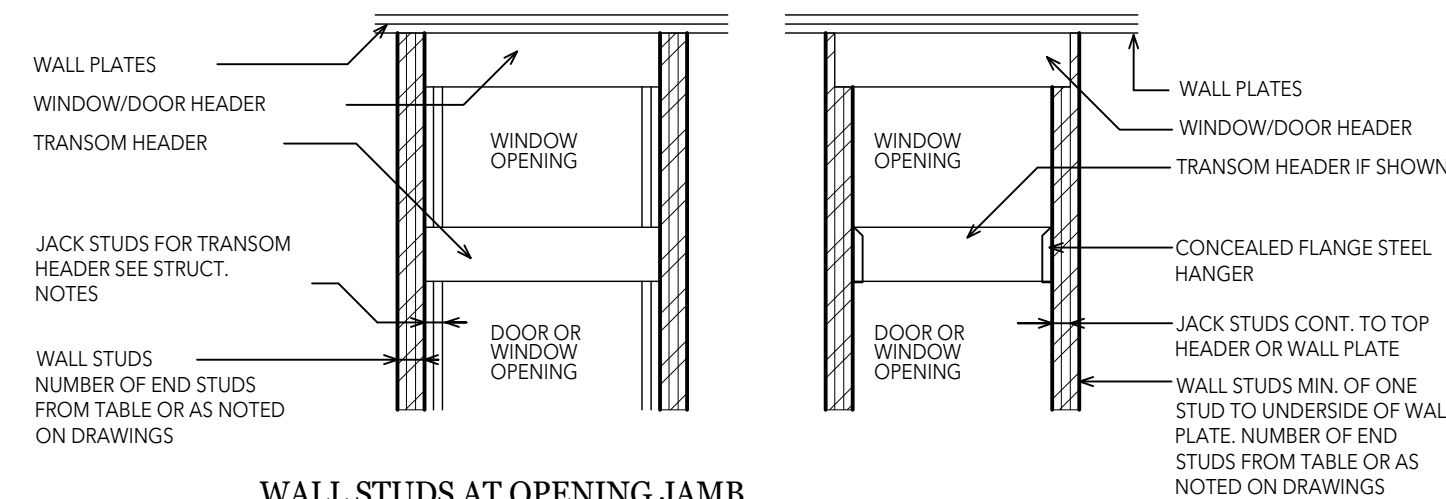
100 PLUMBING TRAP & WATER LINE INSULATION DETAIL
SCALE: 3/4"=1'-0"



WALL ELEVATION
LOAD PATH TIES AT STRUCTURAL SHEATHING



WALL/WINDOW ELEVATION LOAD PATH TIES



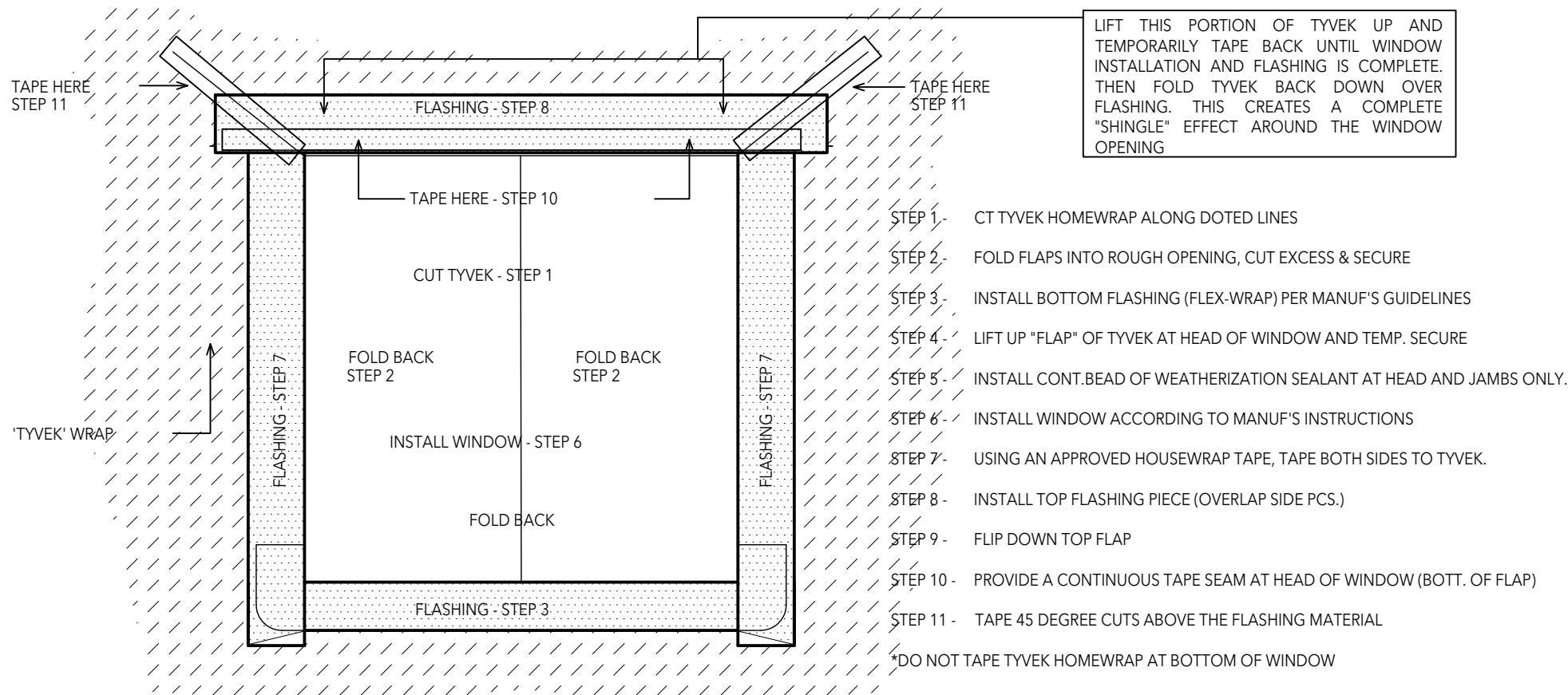
WALL STUDS AT OPENING JAMB

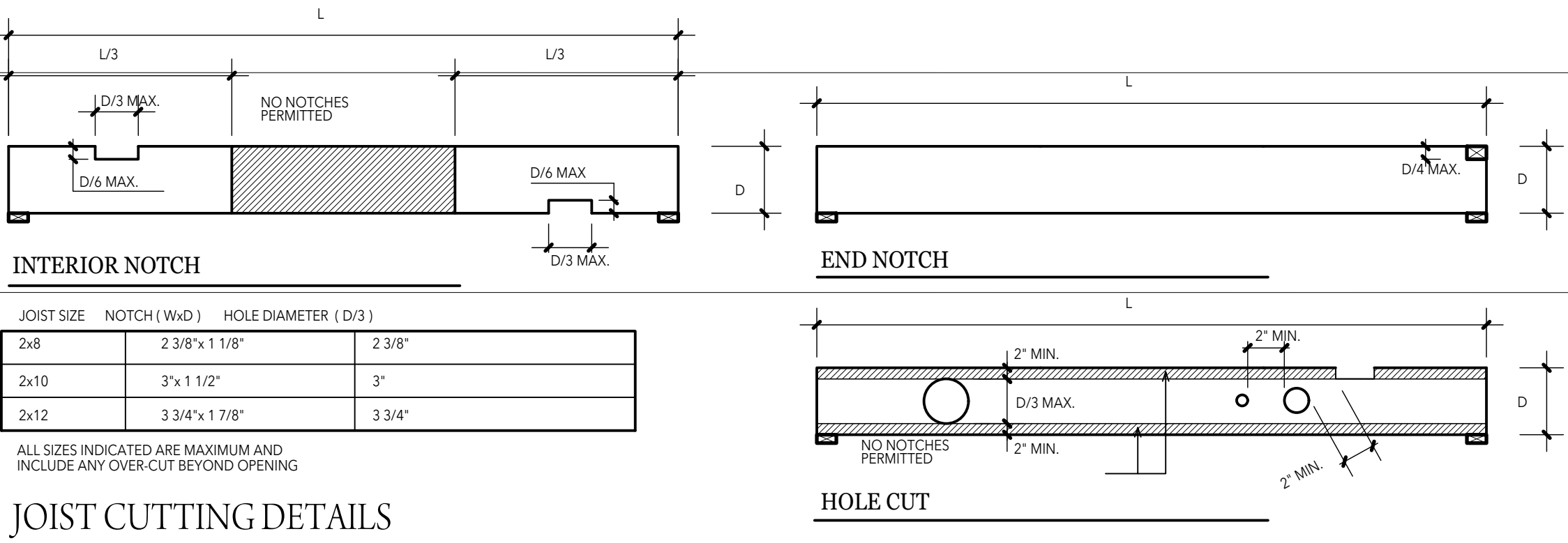
OPENING WIDTH	3'-0"	4'-0"	5'-0"	6'-0"	7'-0"	8'-0"
NUMBER OF STUDS EACH SIDE	1	2	2	3	3	4

*FOR WALL HEIGHTS 10'-0" TO 17'-0"

WINDOW/DOOR END STUD DETAIL

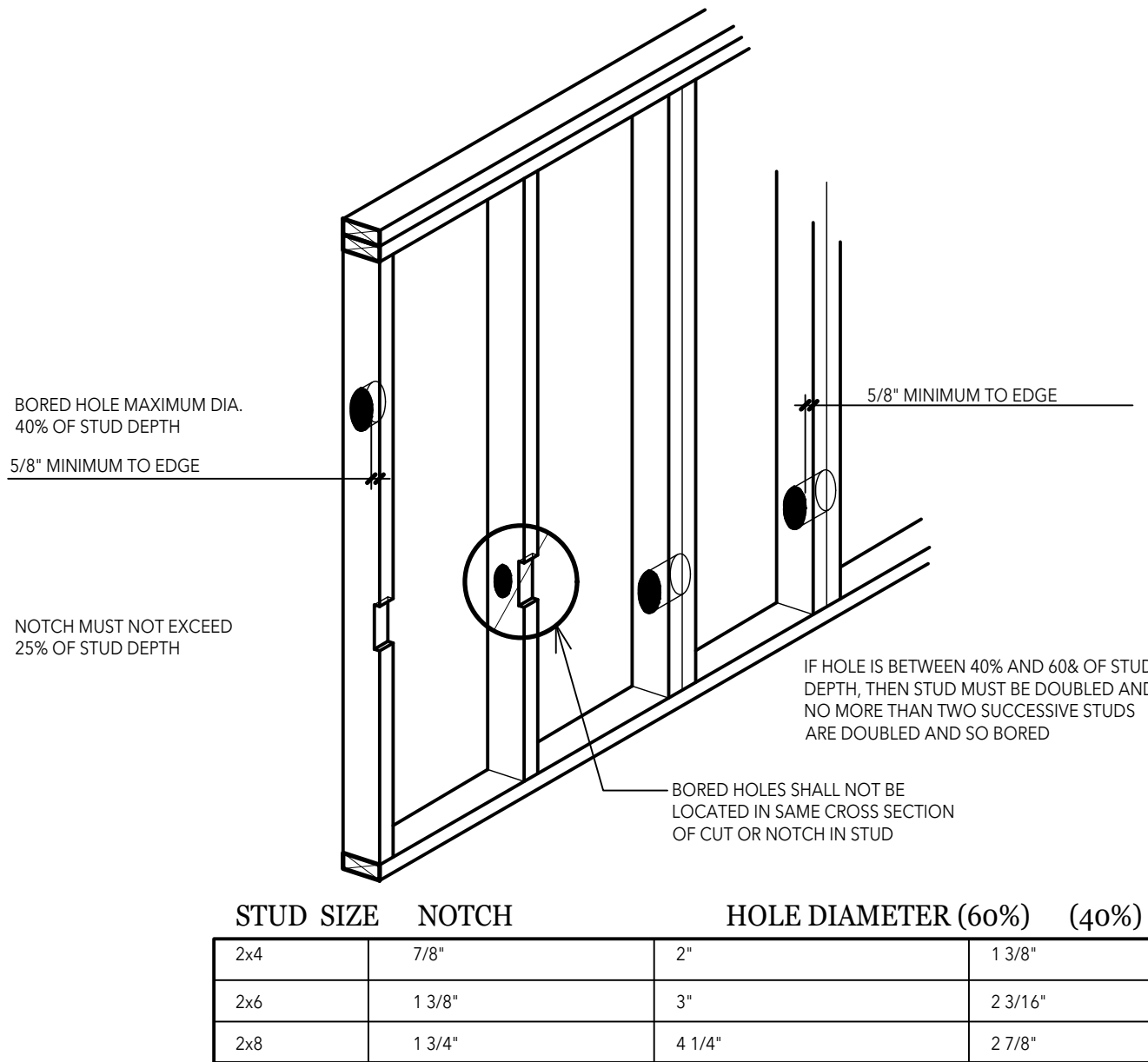
A MINIMUM OF ONE WALL STUD SHALL BE CONTINUOUS TO WALL PLATE AT EACH SIDE OF OPENING



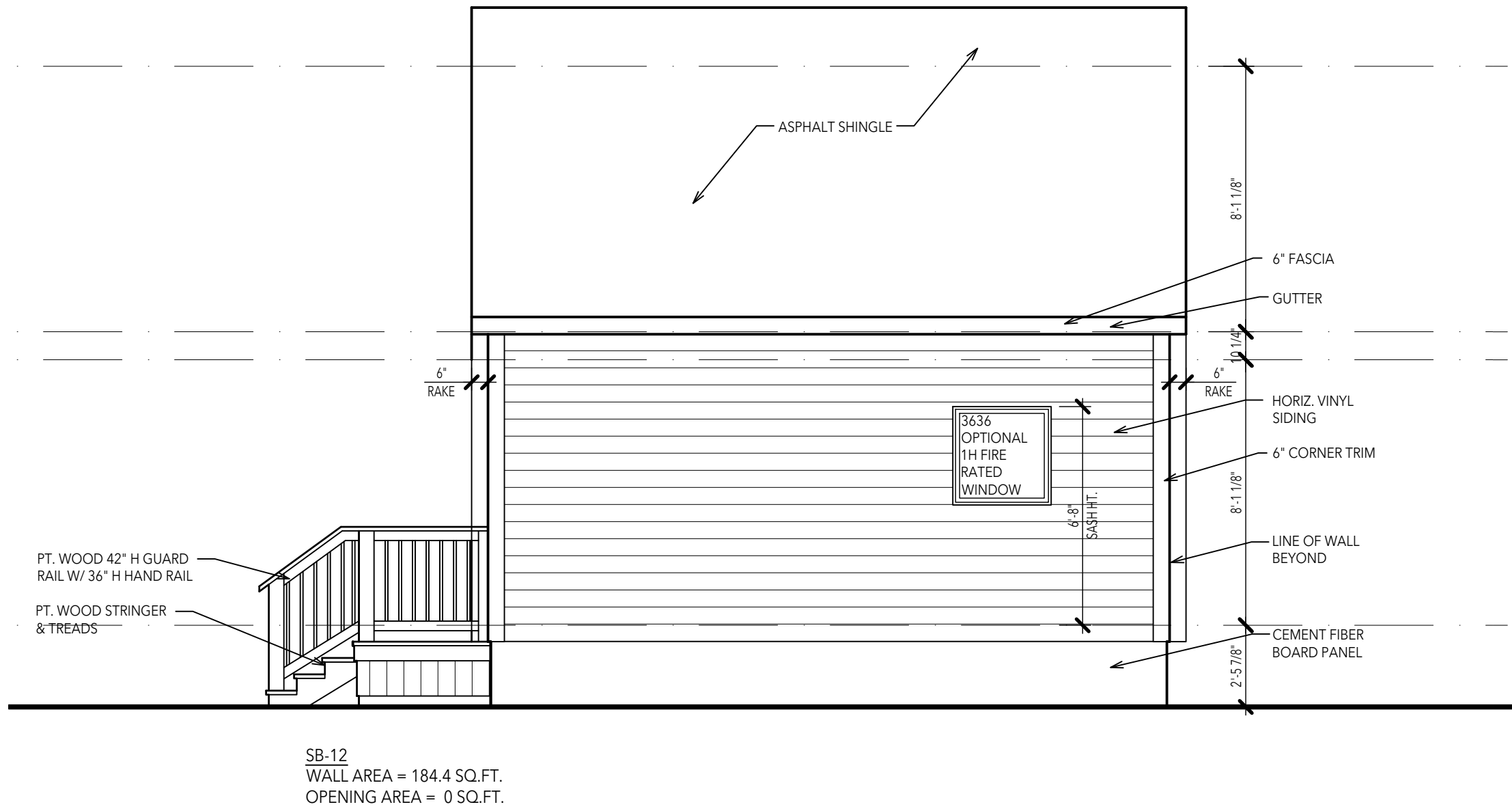


JOIST CUTTING DETAILS

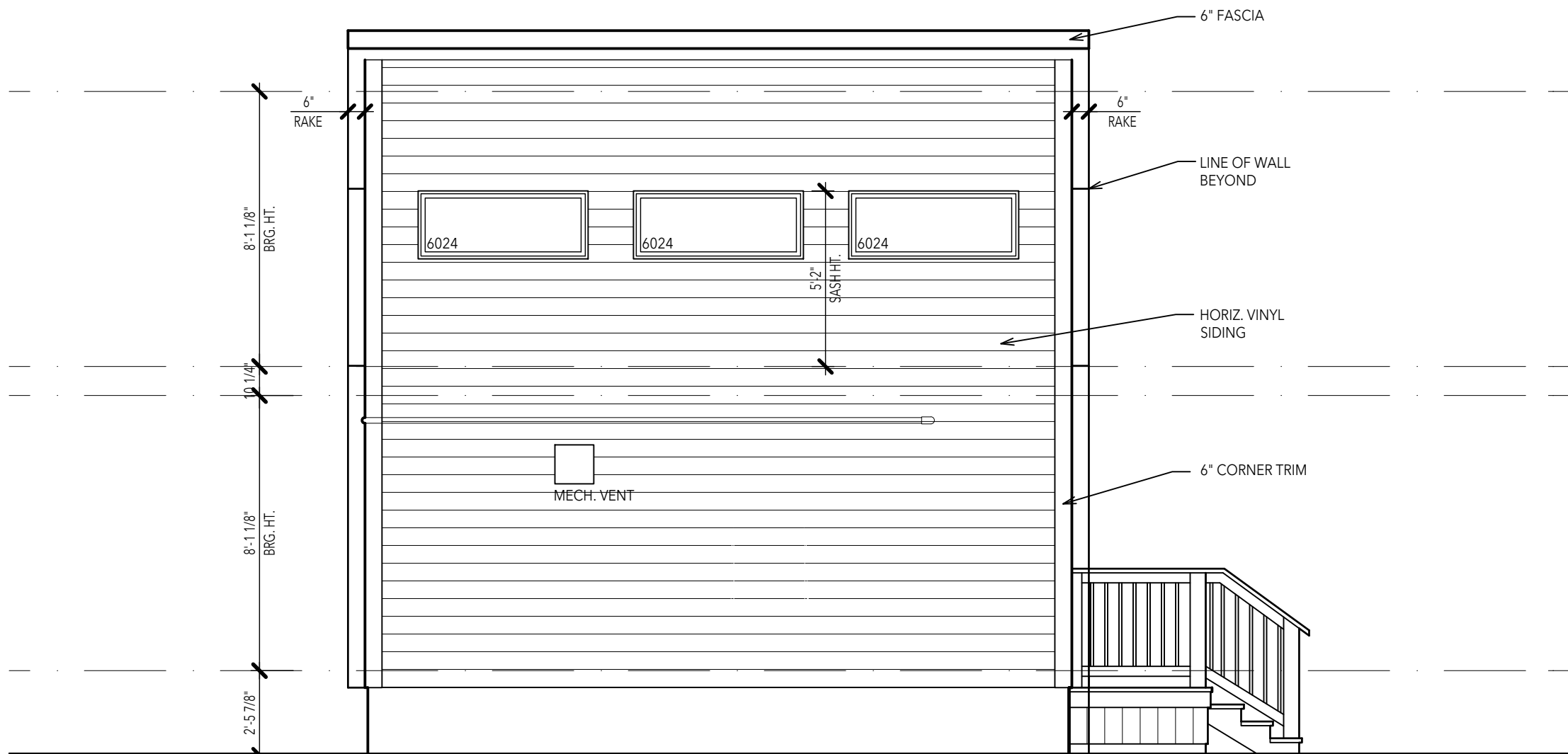
MAXIMUM GLASS AREA (AREAS FOR WHICH 1-IN-50 HOURLY WIND PRESSURE (HWP) IS LESS THAN 0.55kPa ⁽¹⁾)				
GLASS THICKNESS, MM	SHELTERED		OPEN TERRAIN	
	MAXIMUM GLASS AREA, m ² (ft ²)		MAXIMUM GLASS AREA, m ² (ft ²)	
	FACTORY SEALED INSULATED GLASS (IG) UNITS ⁽²⁾	HEAT STRENGTHENED OR TEMPERED GLASS	FACTORY SEALED INSULATED GLASS (IG) UNITS ⁽²⁾	TEMPERED GLASS
2.5	1.02 (11.0)	1.24 (13.4)	0.80 (8.6)	1.25 (13.5)
3	1.71 (18.4)	1.93 (20.8)	1.34 (14.4)	1.74 (18.7)
4	2.68 (28.9)	2.60 (28.0)	2.11 (22.7)	2.33 (25.1)
5	3.74 (40.3)	3.18 (34.2)	2.93 (31.5)	2.86 (30.8)
6	5.24 (56.4)	3.99 (43.0)	4.10 (44.1)	3.59 (38.6)
8	7.93 (85.4)	5.55 (59.7)	6.90 (74.3)	5.00 (53.8)
10	9.92 (106.8)	6.99 (75.2)	9.66 (104.0)	6.26 (67.4)
12	13.92 (149.8)	9.74 (104.8)	12.53 (134.9)	8.78 (94.5)
(1) THE MAXIMUM HOURLY WIND PRESSURE WITH ONE CHANCE IN FIFTY OF BEING EXCEEDED IN ANY ONE YEAR, AS PROVIDED IN MMAH SUPPLEMENTARY STANDARD SB-1, "CLIMATIC AND SEISMIC DATA"				
(2) MAXIMUM GLASS AREA VALUES APPLY TO IG UNITS OF TWO IDENTICAL LITES (ANNEALED, HEAT STRENGTHENED OR TEMPERED) SPACED AT 12.7MM				



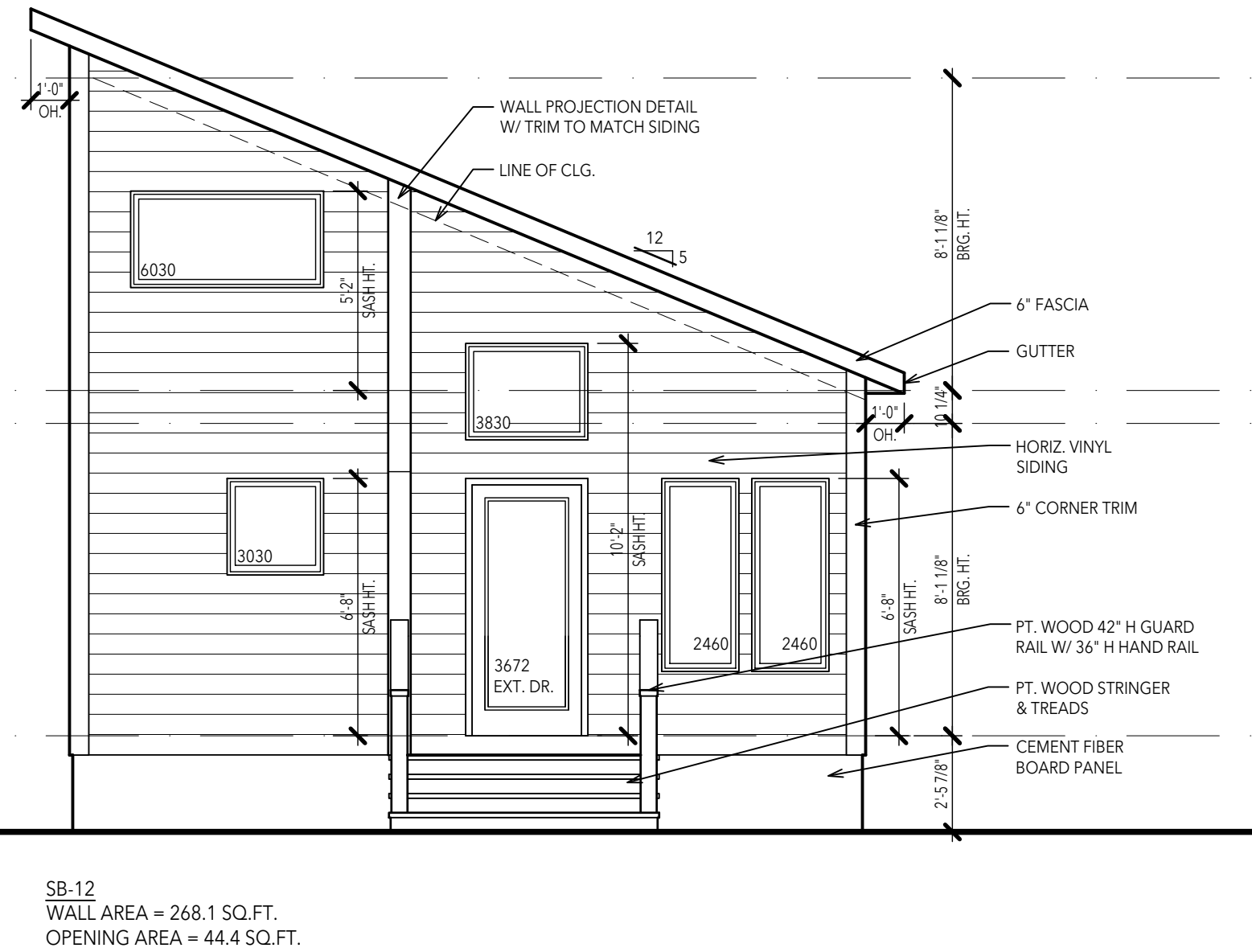
STUD BORING & CUTTING DETAILS



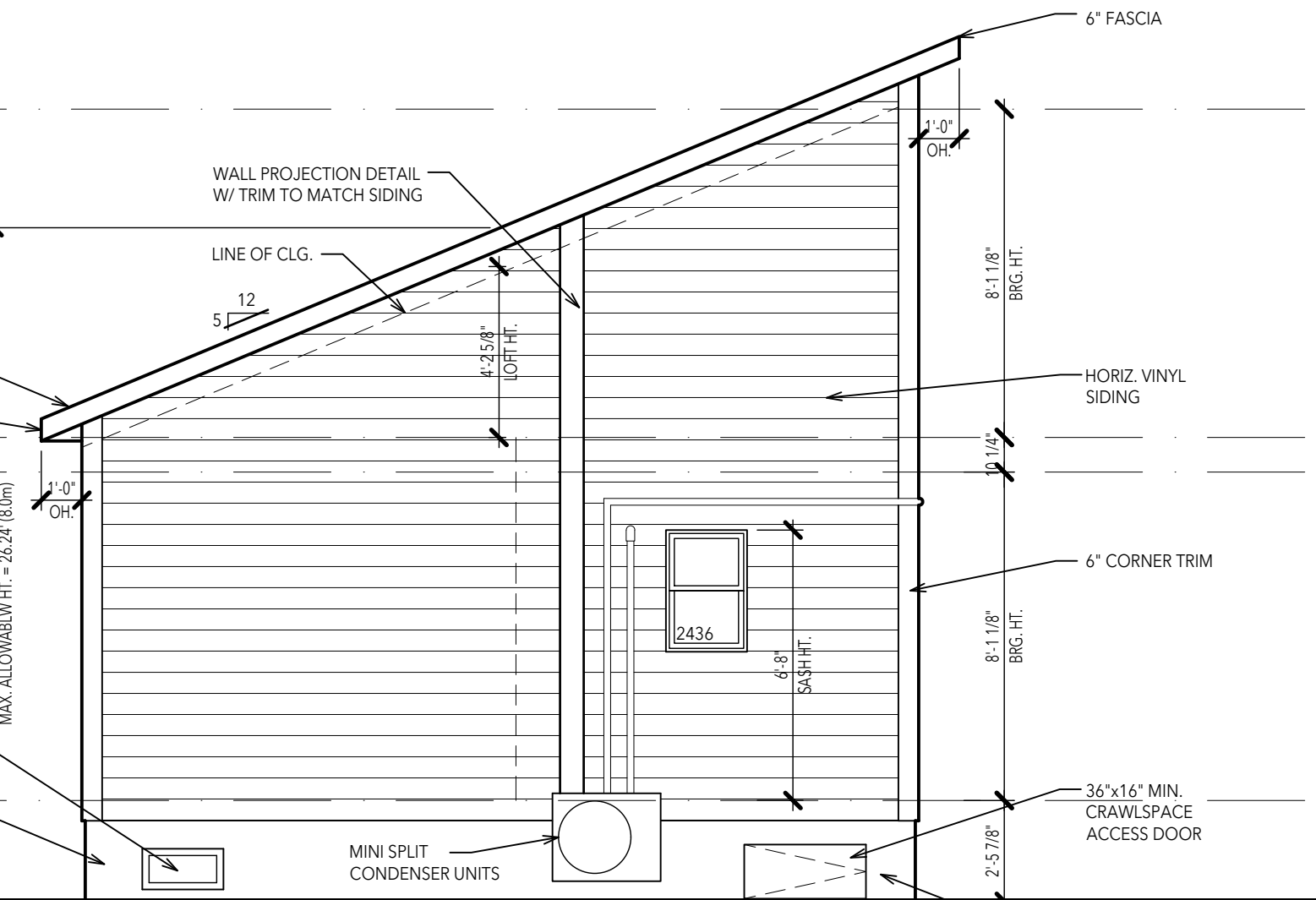
EAST SIDE ELEVATION



WEST SIDE ELEVATION



NORTH SIDE ELEVATION



SOUTH SIDE ELEVATION

DO NOT SCALE THESE DRAWINGS. USE CALCULATED DIMENSIONS. IF VARIATIONS OCCUR CONTACT ARCHITECT FOR CLARIFICATION.

BEFORE CONSTRUCTION OBTAIN WINDOW & DOOR ROUGH OPENING SIZES FROM WINDOW/DOOR SUPPLIER.

NOTE:
ADJUST THE DEPTH OF STONE TO ALLOW FULL BEARING OF MASONRY @ ALL PROJECTED LOCATIONS AS MENTIONED

SEE SHEET #1 FOR ADDITIONAL NOTES & DETAILS REGARDING DRAWINGS & CONSTRUCTION.

ALL INTERIOR PARTITION DIMENSIONS ARE 3 1/2" UNLESS NOTED OTHERWISE.

STEEL LINTEL SCHEDULE	
LOOSE STEEL LINTELS FOR MASONRY - EXTERIOR ANGLES FOR BRICK OR STONE (NO FLOOR LOAD)	
MAX. CLEAR SPAN	LINTEL SIZE
5'-0" OR LESS	3 1/2" x 3 1/2" x 5/16"
7'-0" OR LESS	4" x 3 1/2" x 5/16"
8'-0" OR LESS	5" x 3 1/2" x 5/16"
9'-0" OR LESS	5" x 3 1/2" x 3/8"
10'-0" OR LESS	6" x 3 1/2" x 3/8"
NOTE: THIS SCHEDULE APPLIES UNLESS NOTED OTHERWISE ON THE PLANS AND/OR ELEVATIONS.	

OVERHANGS & DRAINAGE

PROVIDE GUTTERS & DOWNSPOUTS FOR DRAINAGE OF ROOF WATER. DOWNSPOUTS ARE TO BE LOCATED SO THAT THE DISCHARGE WILL NOT SPILL ON OR FLOW ACROSS ANY PORCHES, WALKS OR DRIVES.

UNLESS NOTED OTHERWISE TRUSS OVERHANG DIMENSIONS ARE 1'-0" FROM EXT. FACE OF SHEATHING. PROVIDE 6" RAKE ON GABLE ROOF ENDS

NOTE:
ALL EXTERIOR TRIM BOARDS AND MILLWORK ARE TO BE "AZEK" OR APPROVED EQUAL UNLESS OTHERWISE INDICATED.

NOTE:
OFFSET ALL PLUMBING VENTS TO REAR SLOPE

TYPICAL WINDOW DESIGNATION

NOTE:
GENERAL REFERENCE FOR ROUGH OPENING SIZES ONLY. CONSULT WITH WINDOW MANUFACTURER FOR EXACT WINDOW SIZES, ROUGH OPENINGS AND OTHER REQUIREMENTS.

WINDOW NOTES:

OPERABLE WINDOWS LOCATED W/IN 1'-7" OF FINISHED FLOOR AND EXCEED 6" FROM GROUND OR OTHER SIDE OF THE WINDOW, SHALL BE PROTECTED BY A GUARD OR EQUIPPED W/A MECHANISM TO RESTRICT THE WINDOW FROM OPENING MORE THAN 4".

SEE GENERAL NOTES SHEET FOR ADDITIONAL NOTES.

*EGRESS WINDOWS, MODIFY WINDOW SIZES SHOWN AS REQUIRED TO MEET ALL CODE EGRESS REQUIREMENTS.

1. PILE DESIGN LOADS (UNFACTORED LOADS):

- SEE CCMC REPORT 13691-R FOR PILE TYPE SP3
- MAXIMUM ALLOWABLE LOAD = 125.0 kN (28.2 kips)

a) SOIL ASSUMED TO BE GENERALLY AS FOLLOWS:

SOIL LAYER
• STIFF CLAY OR DENSE SAND

3. PILE INSTALLER TO USE CALIBRATED EQUIPMENT AND CREATE PILE INSTALLATION REPORT (INCLUDING TORQUE AND DEPTH READINGS) AND SUBMIT TO ENGINEER UPON COMPLETION.

MAXIMUM INSTALL TORQUE - 13.6 kN-m (10.0 ft-kip)
MINIMUM INSTALL TORQUE - 6.78 kN-m (5.0 ft-kip)
MINIMUM EMBEDMENT DEPTH - 2.13 m (7.0 ft)

4. IF MINIMUM TORQUE IS NOT ACHIEVED EXTENSIONS SHOULD BE ADDED TO ACHIEVE GREATER EMBEDMENT AND/OR GREATER TORQUE.

5. MINIMUM PILE EMBEDMENT HELIX MUST BE SITUATED BELOW MAXIMUM FROST PENETRATION DEPTH.

6. TOLERANCES:

- PILE TOP TO BE LOCATED WITHIN 40mm (1.5 in) OF CENTER OF APPLIED LOAD.
- PILE TO BE INSTALLED WITHIN 1.5 DEGREES OF VERTICAL.

7. STRUCTURAL STEEL MEMBERS SHALL CONFORM TO:

CSA S16.1-01 (LIMIT STATE DESIGN).
PIPE SHAFT - MINIMUM ASTM A252 GRADE 3 STEEL PIPE, STANDARD SPECIFICATION
FOR WELDED OR SEAMLESS STEEL PIPE PILES.

HELIX PLATE - ASTM A36 OR CSA G40.21 44W HOT ROLLED STRUCTURAL STEEL PLATE.
(NEW MATERIAL ONLY)

8. CORROSION CRITERIA (ASSUMED):

CORROSION RATE = 0.03 mm/year
DESIGN LIFE = 50 years
WALL THICKNESS LOSS OUTSIDE = 1.5 mm

9. WELDING TO CONFORM TO:

WELD TENSILE STRENGTH = 480 MPa (69.6 ksi)
ALL WELDS TO BE 5/16" (7.9) FILLET WELDS UNLESS NOTED OTHERWISE.

10. PILE AND CAP TO BE GALVANIZED. GALVANIZING TO CONFORM ASTM A123

"STANDARD SPECIFICATION FOR ZINC COATINGS ON IRON AND STEEL PRODUCTS".

GALVANIZING OF STEEL SHALL BE DONE AS FOLLOWS:

- SURFACE PREPARATION AND FABRICATION FOR MATERIALS TO BE GALVANIZED SHALL BE COMPLY WITH ASTM A385.
- RECOMMENDED PRACTICE FOR PROVIDING HIGH QUALITY ZINC COATINGS (HOT DIP) AND REFERENCE ASTM A123.

11. FABRICATION SHALL BE CARRIED OUT IN ACCORDANCE WITH STANDARD PRACTICES AND ALL APPLICABLE CODES.

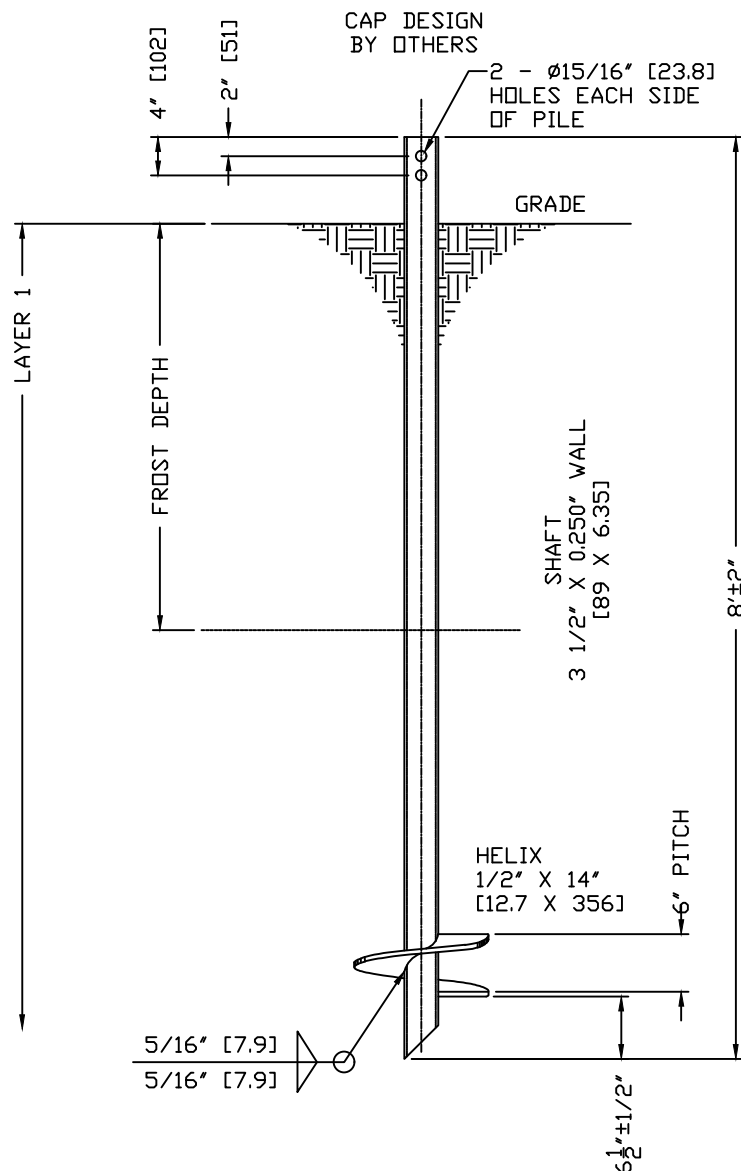
12. ABOVE DESIGN AND COMPONENTS SHALL CONFORM TO THE LATEST EDITIONS OF THE ONTARIO BUILDING CODE, N.B.C. AND APPLICABLE BUILDING CODES.

13. ALL PILES TO BE INSPECTED PRIOR TO INSTALLATION. STRUCTURALLY DAMAGED PILES SHOULD NOT BE INSTALLED.

14. FOR PILE LAYOUT AND LOCATION, REFER TO PHILIP FERNANDES CUSTOM ARCHITECTURAL DESIGN DRAWING A2 OF PROJECT NUMBER 22-1131, DATED NOVEMBER 30, 2022.

NOTE:

- THIS DESIGN AND INFORMATION IS SPECIAL TO ALMITA'S PRODUCT AND IS NON-TRANSFERABLE TO OTHER MANUFACTURERS.



SP3 PILE DETAIL
PIL-0312-250-08-G-001

NOTE:
REFER TO ALMITA DWG. No. 2023-13551B-03
FOR SCREW PILE LOCATION DETAILS



383 ELMIRA ROAD NORTH
GUELPH, ON, CANADA, N1K 1H3
1-800-363-4868
www.almita.com

NOTES

**ISSUED FOR
CONSTRUCTION**



0	18.01.23	IFC	
REV.	DATE-D.M.Y	DESCRIPTION	
PROJECT NUMBER:			
2023-13551B			
CLIENT:			
NC CAPITAL			
PROJECT NAME:			
PRIMARY & DETACHED BUILDINGS			
LOCATION:			
1645 ST LUKE ROAD, WINDSOR, ON			
SHEET TITLE:			
SP3 SCREW PILE DETAILS			
ISSUED FOR:		ISSUE DATE:	SCALE:
CONSTRUCTION		JAN. 18, 2023	NTS
DESIGNED BY:	DRAWN BY:	APPROVED B	
R. SCHMIDT	J. GREWAL	R. SCHMIDT	
DRAWING NUMBER:		REV.:	SHEET:
2023-13551B-02		0	1 OF 1

1. PILE DESIGN LOADS (FACTORED LOADS):

- SEE CCMC REPORT 13691-R FOR PILE TYPE SP2
- MAXIMUM ALLOWABLE LOAD = 111.0 kN (25.0 kips)

a) SOIL ASSUMED TO BE GENERALLY AS FOLLOWS:

SOIL LAYER
• STIFF CLAY OR DENSE SAND

3. PILE INSTALLER TO USE CALIBRATED EQUIPMENT AND CREATE PILE INSTALLATION REPORT (INCLUDING TORQUE AND DEPTH READINGS) AND SUBMIT TO ENGINEER UPON COMPLETION.

MAXIMUM INSTALL TORQUE - 10.8 kN-m (8.0 ft-kip)
MINIMUM INSTALL TORQUE - 2.71 kN-m (2.0 ft-kip)
MINIMUM EMBEDMENT DEPTH - 2.13 m (7.0 ft)

4. IF MINIMUM TORQUE IS NOT ACHIEVED EXTENSIONS SHOULD BE ADDED TO ACHIEVE GREATER EMBEDMENT AND/OR GREATER TORQUE.

5. MINIMUM PILE EMBEDMENT HELIX MUST BE SITUATED BELOW MAXIMUM FROST PENETRATION DEPTH.

6. TOLERANCES:

- PILE TOP TO BE LOCATED WITHIN 40 mm (1.5 in) OF CENTER OF APPLIED LOAD.
- PILE TO BE INSTALLED WITHIN 1.5 DEGREES OF VERTICAL.

7. STRUCTURAL STEEL MEMBERS SHALL CONFORM TO:

CSA S16.1-01 - (LIMIT STATE DESIGN).
PIPE SHAFT - MINIMUM ASTM A252 GRADE 3 STEEL PIPE, STANDARD SPECIFICATION FOR
WELDED OR SEAMLESS STEEL PIPE PILES.
HELIX PLATE - ASTM A36 OR CSA G40.21 44W HOT ROLLED STRUCTURAL STEEL PLATE.
(NEW MATERIAL ONLY)

8. CORROSION CRITERIA (ASSUMED):

CORROSION RATE = 0.03 mm/year
DESIGN LIFE = 50 years
WALL THICKNESS LOSS OUTSIDE = 1.5 mm

9. WELDING TO CONFORM TO:

CSA W59 AND CSA W47.1
WELD TENSILE STRENGTH = 480 MPa (69.6 ksi)
ALL WELDS TO BE 5/16" (7.9 mm) FILLET WELDS UNLESS NOTED OTHERWISE.

10.PILE AND CAP TO BE GALVANIZED. GALVANIZING TO CONFORM ASTM A123

"STANDARD SPECIFICATION FOR ZINC COATINGS ON IRON AND STEEL PRODUCTS".

GALVANIZING OF STEEL SHALL BE DONE AS FOLLOWS:

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- RECOMMENDED PRACTICE FOR PROVIDING HIGH QUALITY ZINC COATINGS (HOT DIP) AND REFERENCE ASTM A123.

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12.ABOVE DESIGN AND COMPONENTS SHALL CONFORM TO THE LATEST EDITIONS OF THE ONTARIO BUILDING CODE, N.B.C. AND APPLICABLE BUILDING CODES.

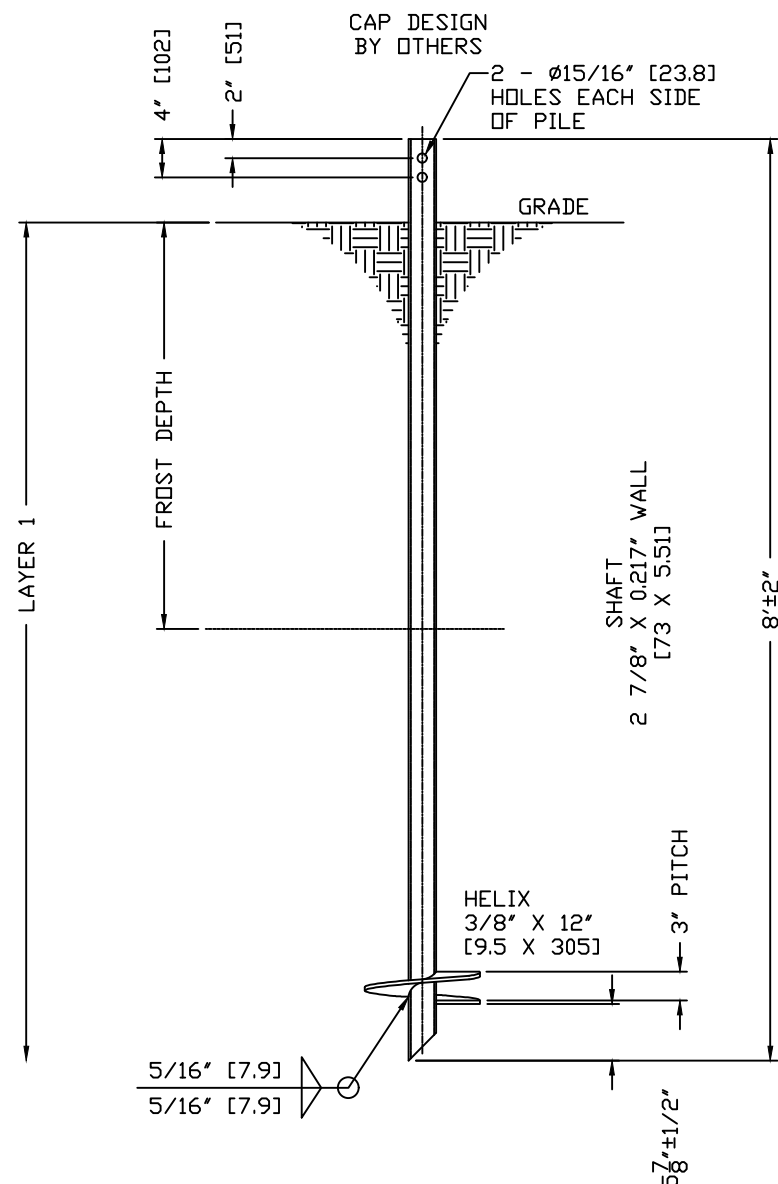
13. ALL PILES TO BE INSPECTED PRIOR TO INSTALLATION. STRUCTURALLY DAMAGED PILES SHOULD NOT BE INSTALLED.

14. FOR PILE LAYOUT AND LOCATION, REFER TO PHILIP FERNANDES CUSTOM ARCHITECTURAL DESIGN DRAWING A2 OF PROJECT NUMBER 22-1131, DATED NOVEMBER 30, 2022.

NOTE:

- THIS DESIGN AND INFORMATION IS SPECIAL TO ALMITA'S PRODUCT AND IS NON-TRANSFERABLE TO OTHER MANUFACTURERS.

NOTE:
REFER TO ALMITA DWG. No. 2023-13551B-03
FOR SCREW PILE LOCATION DETAILS



SP2 PILE DETAIL
PIL-0278-217-08-G-004



383 ELMIRA ROAD NORTH
GUELPH, ON, CANADA, N1K 1H3
1-800-363-4868
www.almita.com

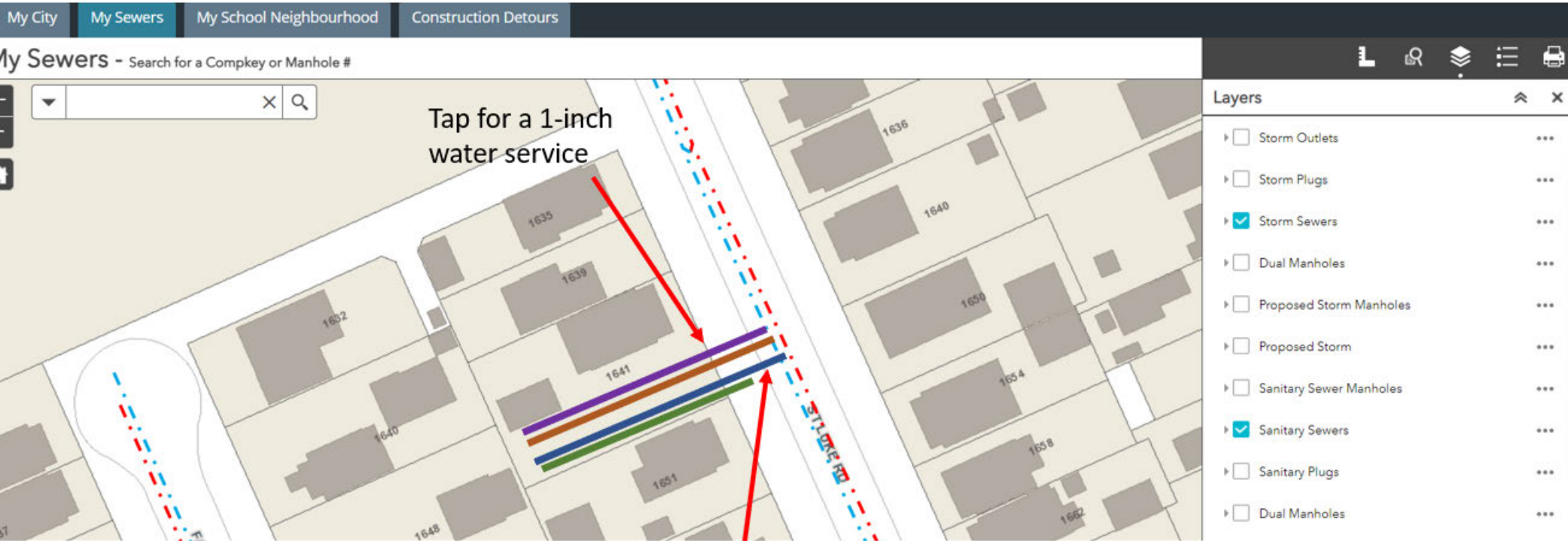
NOTES

**ISSUED FOR
CONSTRUCTION**



0	18.01.23	IFC	
REV.	DATE-D.M.Y	DESCRIPTION	
PROJECT NUMBER:			
2023-13551B			
CLIENT:			
NC CAPITAL			
PROJECT NAME:			
PRIMARY & DETACHED BUILDINGS			
LOCATION:			
1645 ST LUKE ROAD, WINDSOR, ON			
SHEET TITLE:			
SP2 SCREW PILE DETAILS			
ISSUED FOR:		ISSUE DATE:	SCALE:
CONSTRUCTION		JAN. 18, 2023	NTS
DESIGNED BY:	DRAWN BY:	APPROVED BY:	
R. SCHMIDT	J. GREWAL	R. SCHMIDT	
DRAWING NUMBER:		REV.:	SHEET:
2023-13551B-01		0	1 OF 1

1645 St Luke Site Services Sketch



Water Sanitary Storm

Hydro Overhead

Ex. 8 inch pvc
watermain

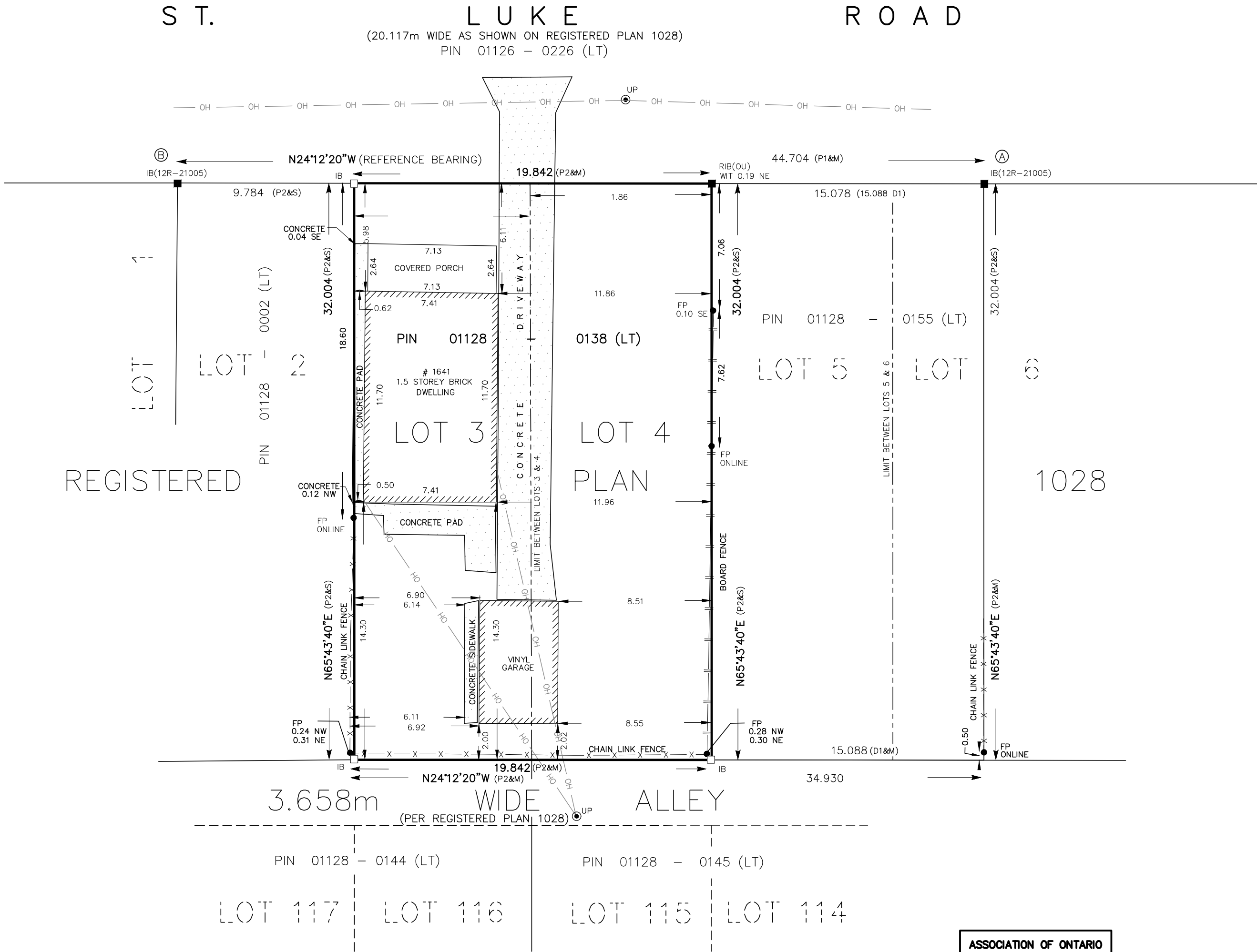
COORDINATES BELOW ARE DERIVED FROM NETWORK RTK GPS OBSERVATIONS (TOPNET) AND ARE REFERRED TO UTM ZONE 17, NAD83 (CSRS) (2010.0).		
COORDINATES COMPLY WITH THE URBAN ABSOLUTE ACCURACY PER SEC. 14(2) OF O.REG. 216/10.		
POINT ID	NORTHING	EASTING
(A)	4685895.08	335579.99
(B)	4685935.87	335561.65
COORDINATES CANNOT, IN THEMSELVES, BE USED TO RE-ESTABLISH CORNERS OR BOUNDARIES SHOWN ON THIS PLAN.		

SURVEYOR'S REAL PROPERTY REPORT – PART 1
PLAN OF SURVEY OF
ALL OF LOTS 3 AND 4
REGISTERED PLAN 1028
CITY OF WINDSOR



PART 2
THIS PLAN MUST BE READ IN CONJUNCTION WITH A SURVEY REPORT DATED JUNE 04, 2021.

THIS REPORT WAS PREPARED FOR JOSEPH BENOIT GENIER & MONIA GENIER AND THE UNDERSIGNED ACCEPTS NO RESPONSIBILITY FOR USE BY OTHER PARTIES.



LEGEND	
□	MONUMENT PLANTED
■	MONUMENT FOUND
IB	IRON BAR
RIB	ROUND IRON BAR
M	MEASURED
S	SET
P1	PLAN 12R-21005
P2	REGISTERED PLAN 1028
D1	INSTRUMENT NUMBER R1182537
D2	INSTRUMENT NUMBER R1259454
OU	ORIGIN UNKNOWN
FP	FENCE POST
UP	UTILITY POLE
—OH—	OVERHEAD SERVICES

NOTES

DISTANCES AND COORDINATES SHOWN ON THIS PLAN ARE IN METRES AND CAN BE CONVERTED TO FEET BY DIVIDING BY 0.3048.

DISTANCES ON THIS PLAN ARE HORIZONTAL GROUND DISTANCES AND CAN BE CONVERTED TO GRID DISTANCES BY MULTIPLYING BY THE AVERAGE COMBINED SCALE FACTOR OF 0.999892.

BEARINGS ARE UTM GRID DERIVED FROM NETWORK RTK GPS OBSERVATIONS (TOPNET) ON MONUMENTS (A) AND (B) AS SHOWN HEREON, HAVING A GRID BEARING OF N24°12'20"W, NAD83 CSRS (2010.0) AND ARE REFERRED TO THE CENTRAL MERIDIAN OF UTM ZONE 17 (81° W LONGITUDE).

ALL BUILDING TIES SHOWN HEREON ARE TO THE FACE OF THE SIDING UNLESS OTHERWISE NOTED.

FOR BEARING COMPARISONS, THE FOLLOWING ROTATIONS WERE APPLIED:
P1 – 0°58'35" CLOCKWISE.
P2 – 1°19'20" CLOCKWISE.

SURVEYOR'S CERTIFICATE

I CERTIFY THAT:

1. THIS SURVEY AND PLAN ARE CORRECT AND IN ACCORDANCE WITH THE SURVEYS ACT, THE SURVEYORS ACT, THE LAND TITLES ACT AND THE REGULATIONS MADE UNDER THEM.

2. THE SURVEY WAS COMPLETED ON THE 04th DAY OF MAY, 2021.

JUNE 04, 2021

MICHAEL W. HAINES
ONTARIO LAND SURVEYOR

© THIS PLAN IS PROTECTED BY COPYRIGHT. NO PERSON MAY COPY, REPRODUCE, OR ALTER THIS PLAN IN WHOLE OR IN PART WITHOUT THE WRITTEN AUTHORIZATION OF SURVEYORS ON SITE INC.

ASSOCIATION OF ONTARIO
LAND SURVEYORS
PLAN SUBMISSION FORM
2167509

THIS PLAN IS NOT VALID UNLESS IT IS AN EMBOSSED ORIGINAL COPY ISSUED BY THE SURVEYOR in accordance with Regulation 1026, Section 29(3).

CLARKE SOS A DIVISION OF SURVEYORS ON SITE INC.	Ontario Land Surveyors Consulting Surveyors 2535 Lesperance Road Tecumseh, Ontario N8N 2X1 Ph. (519) 258-4166 Toll Free: 1-888-735-4166 www.surveyorsonsite.com	DRAWN BY HK
		CHECKED BY MWH
		JOB NO. 2021-253
		FILE 2021-253-SRPR
		CAD FILE L-1028-0

Application for a Permit to Construct or Demolish

Project information				
Unit number 1645	Lot / con. Lot 4	Postal code N8Y 3N4		
Plan number / other description LT 4 PL 1028 FORD CITY; WINDSOR				
Estimated cost of construction 150000 CAD	Area of work 40.0 m ²			
Purpose of application				
Current use of building Vacant Land				
Description of proposed work Main building is a 2-storey vinyl siding 2-unit, each unit 2bd 1bth 670 SF. Rear building is a 1 storey detached approx 431 SF 1bd 1bth adu with a shed-style roof				

Designer Information

Individual who reviews and takes responsibility for design activities

Last name Fernandes	First name Philip	Corporation or partnership
Street address 956 McDougall St	Unit number	Lot / Con.
Municipality Windsor	Postal code N9A 1M1	Province Ontario
Other phone	Mobile phone +1 (519) 254-4949	
Fax	Email philip@pfernandesdesigns.com	

Design activities undertaken by the Designer

Select one or more design activities

- | | | |
|---|---|---|
| ☐ House | ☒ Small buildings | ☐ Large buildings |
| ☐ Complex buildings | ☐ HVAC - House | ☐ Building services |
| ☐ Detection, lightning and power | ☐ Fire protections | ☐ Building structural |
| ☐ Plumbing - house | ☐ Plumbing - All buildings | ☐ On-site sewage systems |

Description of designer's work

ADU

Designer Qualification and Registration

Qualification and registration

Designer is qualified and the company is registered

Individual BCIN

37630

Company BCIN

47050

Certification of the Designer

I certify that the information contained in this schedule is true to the best of my knowledge. I have submitted this application with the knowledge and consent of the firm.

By signing off, I understand that it constitutes a legal signature confirming that I acknowledge and agree to the above declaration.



Signed by **Philip Fernandes** December 22, 2022 at 3:59:56 p.m. EST

Designer Information

Individual who reviews and takes responsibility for design activities

Last name Fernandes	First name Philip	Corporation or partnership
Street address 956 McDougall St	Unit number	Lot / Con.
Municipality Windsor	Postal code N9A 1M1	Province Ontario
Other phone	Mobile phone +1 (519) 254-4949	
Fax	Email philip@pfernandesdesigns.com	

Design activities undertaken by the Designer

Select one or more design activities

- | | | |
|---|---|---|
| ☐ House | ☒ Small buildings | ☐ Large buildings |
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By signing off, I understand that it constitutes a legal signature confirming that I acknowledge and agree to the above declaration.



Signed by **Philip Fernandes** December 22, 2022 at 3:59:56 p.m. EST

Energy Efficiency Design Summary

Energy Efficiency Compliance Method

Compliance method Prescriptive	Compliance package 3.1.1.2.A Package - A1		
-----------------------------------	--	--	--

Project Design Conditions

Climatic zone Zone 1 (< 5000 degree days)	Space heating fuel source Electric	Heating equipment efficiency ≥ 92% AFUE
Area of walls 99.86 m ²	Area of windows, skylights and glass 8.621 m ²	Area ratio 8.64 %
Utilize window averaging? ☐ Yes ☒ No		

Other building characteristics

- | | | |
|--|--|---|
| ☐ Log/Post & beam | ☐ Slab-on-ground | ☐ Air conditioning |
| ☒ Air sourced heat pump (ASHP) | ☐ Ground sourced heat pump (GSHP) | ☐ ICF above grade |
| ☐ Walkout basement | ☐ Combo unit | ☐ ICF basement |

Energy Efficiency Substitutions

Energy efficiency substitutions

- | | | |
|--|---|--|
| ☐ ICF (3.1.1.2.(5)&(6)/3.1.1.3.(5)&(6)) | ☐ Airtightness substitution(s) | ☐ Combined space heating and domestic water heating systems (3.1.1.2.(7)/3.1.1.3.(7)) |
|--|---|--|

Thermal Insulation

Building component	Min. nominal	Min. effective	Max U
Ceiling with attic space	60	59.22	0
Ceiling without attic space	31	27.65	0
Exposed floor	31	29.8	0
Walls above grade	22	29.8	0
Basement walls	0	0	0
Below grade slab entire surface > 600 mm below grade	0	0	0
Heated slab or slab ≤ 600 mm below grade	0	0	0
Edge of below grade slab ≤ 600 mm below grade	0	0	0

Windows and Doors		
Building component	Energy rating	Max U
Windows/Sliding glass doors	25	0.28
Skylights/Glazed roofs		0.48

Mechanicals	
Building component	Energy rating
Space heating equipment (min. AFUE)	96
HRV efficiency (SRE% at 0°C)	75
Domestic water heater (EF)	0.8
DWHR (CSA B55.1 (min. 42% efficiency))	42
Combined heating system	
Amount of showers 1	

Declaration	
I hereby declare that I have reviewed and take responsibility for the design work.	
	Signed by Philip Fernandes January 13, 2023 at 9:08:57 a.m. EST
	Signed by Philip Fernandes December 22, 2022 at 4:06:17 p.m. EST Note: Form has been edited after the sign off.

Residential Mechanical Ventilation and HVAC Design Summary

Heating and Combustion	
House Type IV Type I or II with electric space heat	Heating system Electric Space Heating
Heating fuel type ☐ Gas ☐ Oil ☐ Propane ☒ Electric	Combustion Appliances ☐ No combustion appliances ☒ Direct vent (sealed combustion) only ☐ Natural draft, B vent or induced draft fireplace ☐ Positive venting incuded draft (excluding fireplace) ☐ Solid fuel (including fireplace)

Ventilation				
Ventilation capacity		Range Hood Vented to Exterior ☒ Yes		
Description		Multiplier	Air flow	
Master Bedroom & Basement	1	10	10.0 L/s	
Other bedrooms	0	5	0 L/s	
Bathroom & Kitchen	2	5	10.0 L/s	
Other rooms	2	5	10.0 L/s	
Total ventilation capacity			30.0 L/s	
Less principal ventilation capacity	1 Bedroom		15.0 L/s	
Required supplemental ventilation capacity			15.0 L/s	
Heat Recover Ventilator (HRV)				
Make/Model Lifebreath 195 DCS		HVI Certified ☒ Yes		
Air flow High 87.31 L/s		Air flow Low 33.98 L/s		
Sensible Efficiency @ 0°C (%) 81 %	Sensible Efficiency @ 0°C - (w)	Sensible Efficiency @ -25°C (%) 68 %	Sensible Efficiency @ -25°C (w)	
NOTE! Where a Heat Recovery Ventilator (HRV) is connected to a forced air heating system, the HRV supply shall be directly connected to the forced air heating system return air side.				
Supplemental Fan Capacities				
Location	Make	Model	Design Airflow	Sones
Bath	NuTone	ARN90C	42.475 L/s	2
NOTE! The principal exhaust fan shall be controlled by a centrally located manual switch and labelled VENTILATION FAN.				

Appliances

Heating Appliance

Make/Model Mitsubishi MXZ-3C24NA (or equivalent)	Heating Output 7.327 kW	Total Design Heat Load 5.198 kW
---	----------------------------	------------------------------------

Cooling Appliance

Make/Model Mitsubishi MXZ-3C24NA (or equivalent)	Cooling Output 6.448 kW	Total Design Cooling Load 3.626 kW
---	----------------------------	---------------------------------------

Certification

I hereby certify that this ventilation and heating system has been designed in accordance with the Ontario Building Code



Signed by **Lawrence Mailloux** January 2, 2023 at 3:07:47 p.m. EST

Supplemental Information for the Application

Area and Height

Gross floor area

40.0 m²

Finished basement area

0.0 m²

Amount of Units/Stories

Amount of dwelling units created

1

Total amount of dwelling units

1

Amount of stories

1

Deck and Garage

Is there a deck or garage attached?

☐ Deck

☐ Garage